CSE 4303
Data Structures

Topic: Topological Sort

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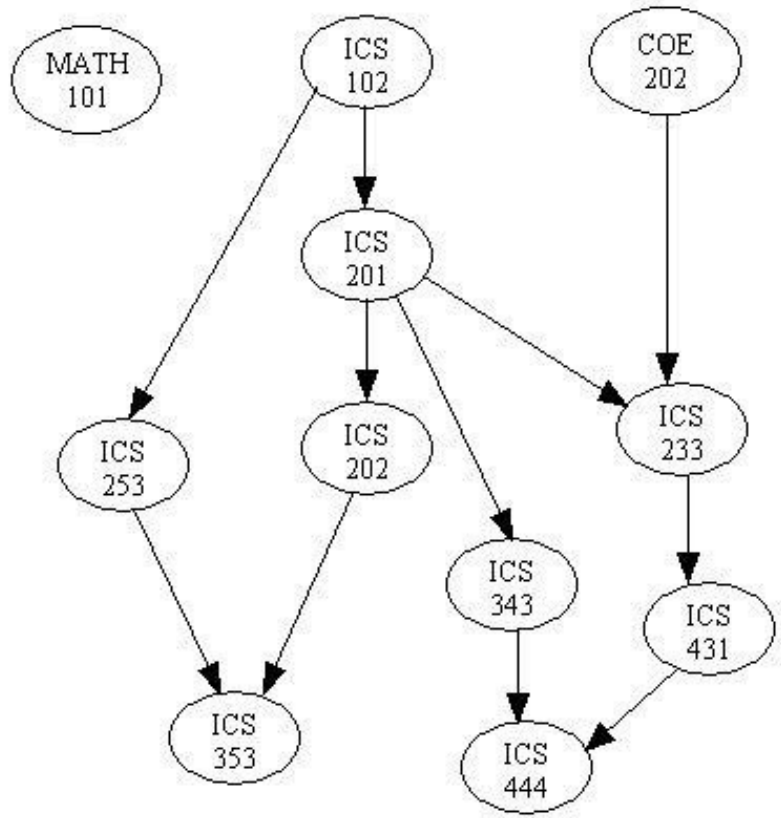
Motivation

- Given a set of **tasks with dependencies**:
  - Find is there an **order** in which we can complete the tasks?

- Cycles in dependencies can cause issues…

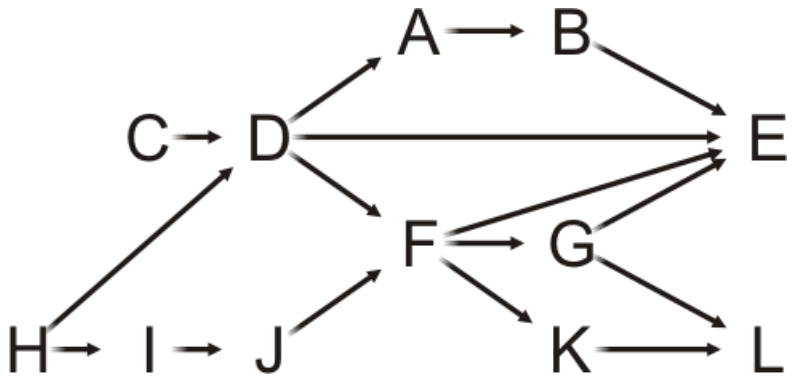


Definition

- Topological sorting of the vertices in a DAG:
  - An ordering $v_1, v_2, v_3, \ldots, v_{|V|}$ such that v_j appears before v_k if there is a path from v_j to v_k.

- Given this DAG, a topological sort is

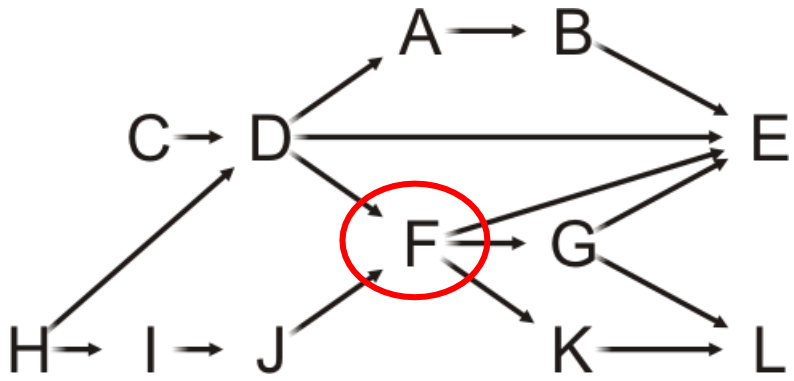
$$H, C, I, D, J, A, F, B, G, K, E, L$$



Example

- There are paths from H, C, I, D, J to F, so all these must come before F in a topological sort

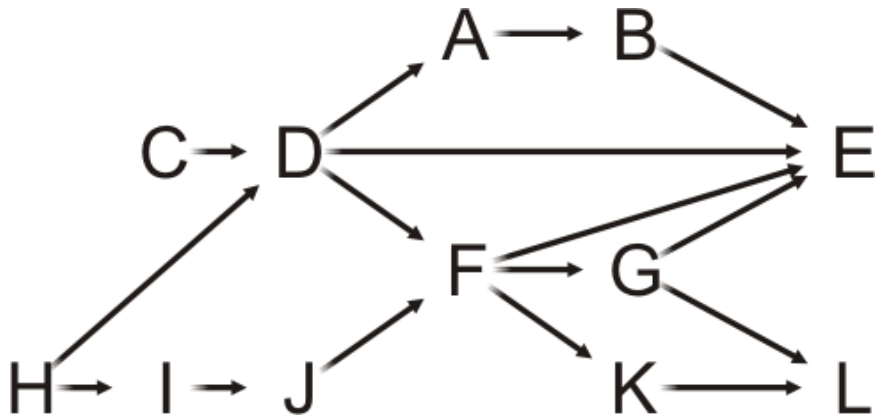
$$\boldsymbol{H, C, I, D, J}, A, \boldsymbol{F}, B, G, K, E, L$$



Clearly, this sorting may not be unique

Topological Sort

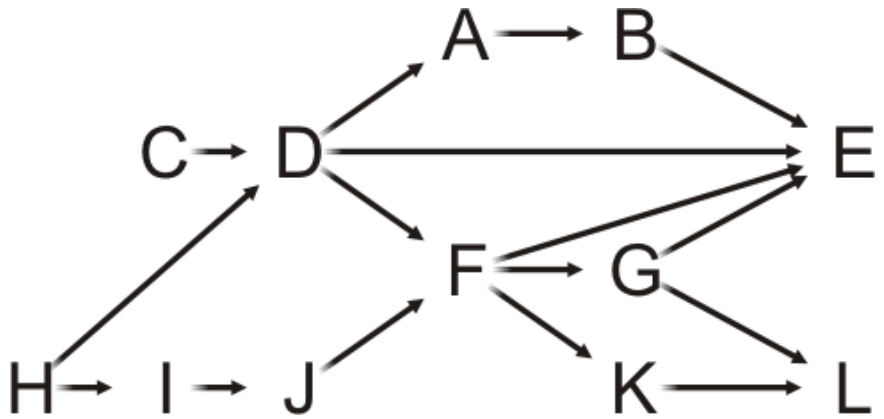
- **Idea**:
- Given a DAG V, make a copy W and iterate:
  - Find a vertex v in W with **in-degree zero**
  - Let v be the next vertex in the topological sort
  - Continue iterating with the vertex-induced sub-graph $W \backslash \{v\}$



Topological Sort

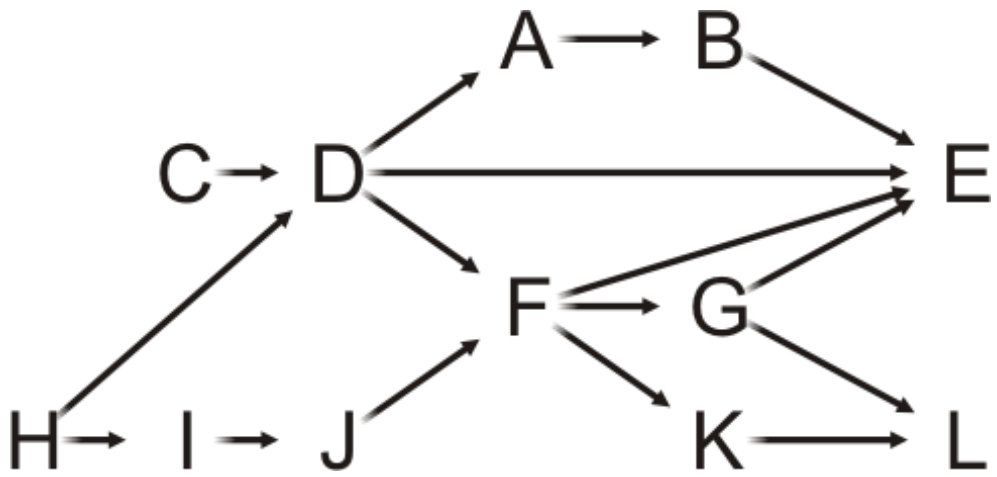
- **Idea**:
- Given a DAG V, make a copy W and iterate:
  - Find a vertex v in W with **in-degree zero**
  - Let v be the next vertex in the topological sort
  - Continue iterating with the vertex-induced sub-graph $W \backslash \{v\}$

- On this graph, iterate the following $|V| = 12$ times
  - Choose a vertex ***v* that has in-degree zero**.
  - Let v be the next vertex in our topological sort
  - Remove v and all edges connected to it



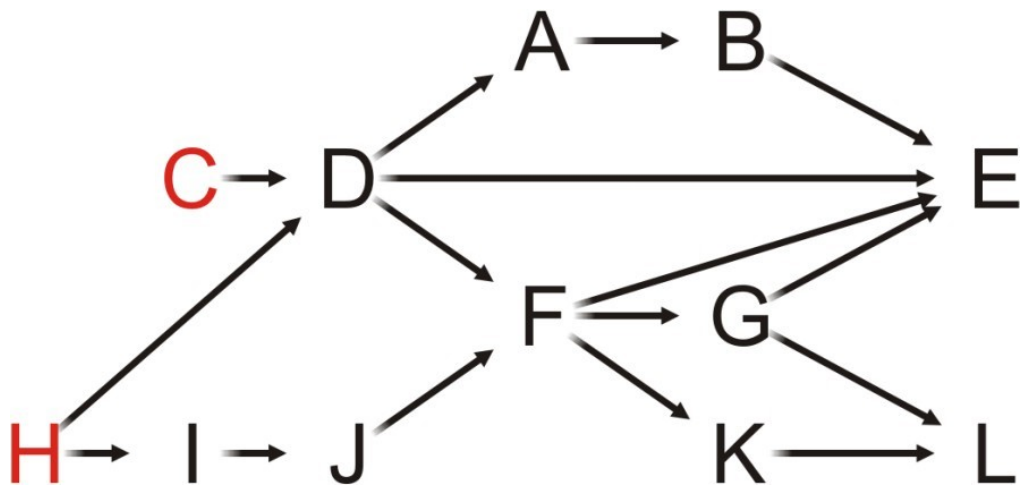
Example

- Let’s step through this algorithm with this example
  - Which task can we start with?

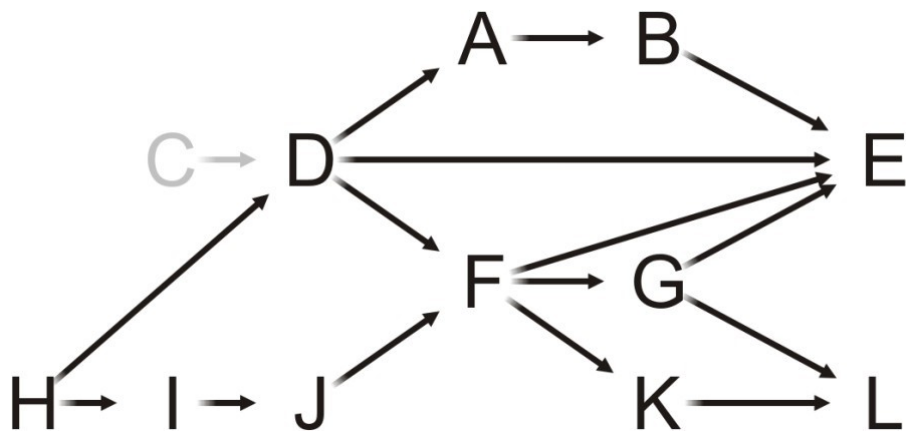


- Out of Tasks C or H, let's choose Task C

  (we could also start with H, hence Topological order need not be unique)

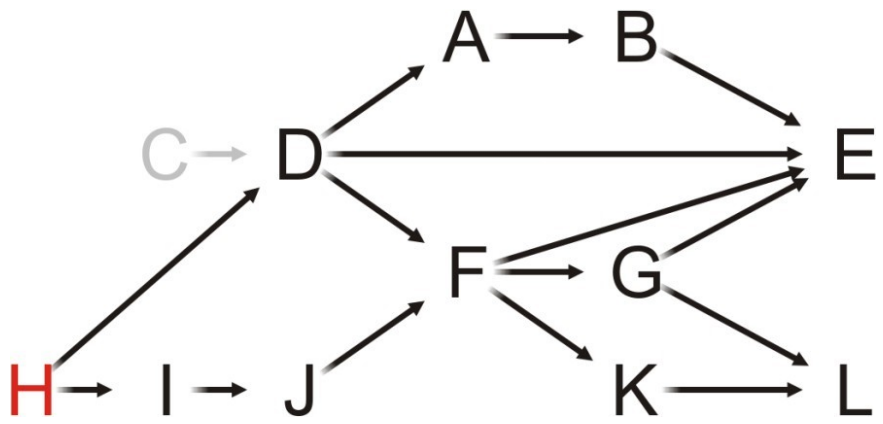


- Having completed Task C, which vertices have in-degree zero?



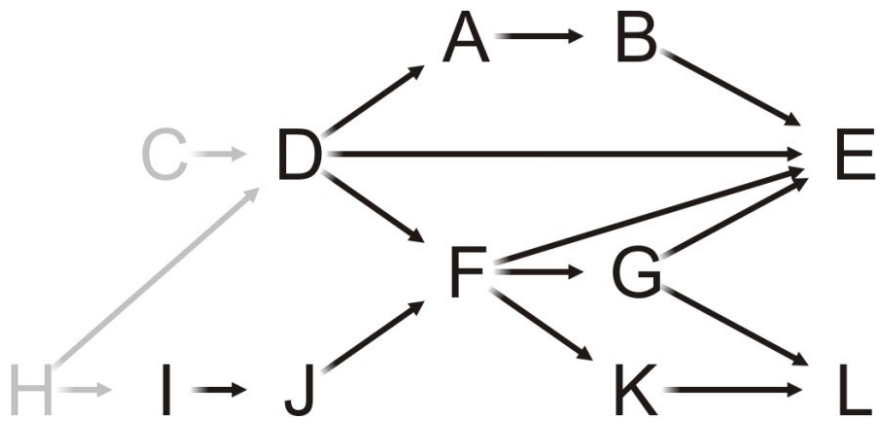
C

- Only Task H can be completed, so we choose it



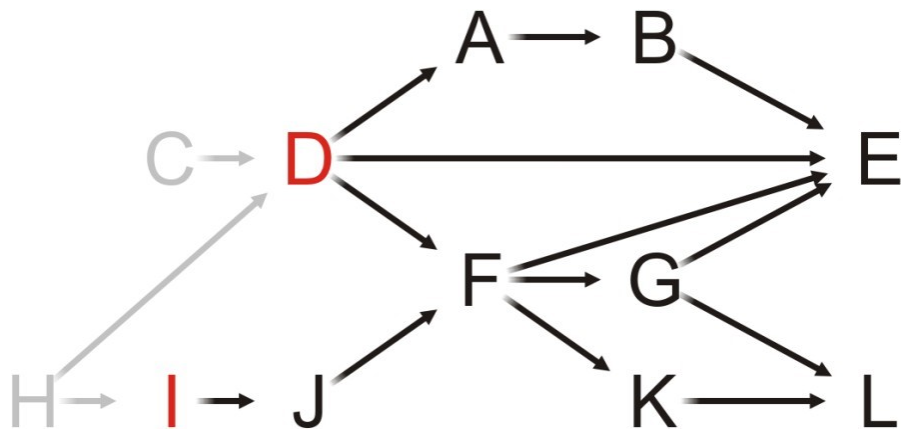
C

- Having removed H, what is next?



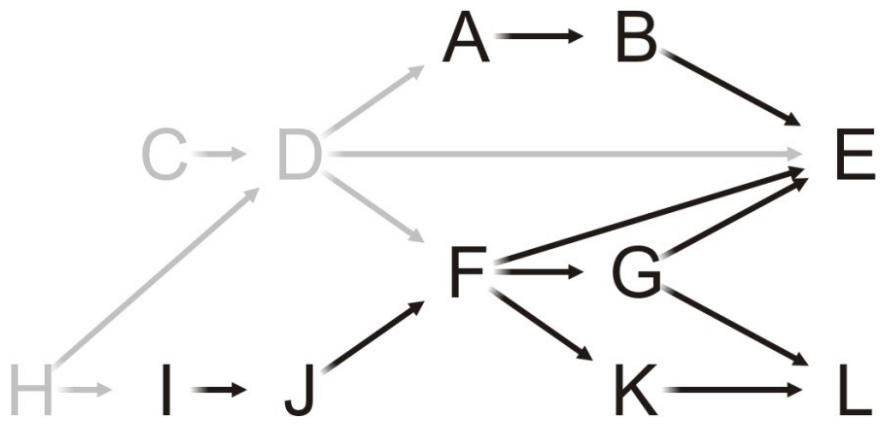
C, H

- Both Tasks D and I have in-degree zero
- Let us choose Task D



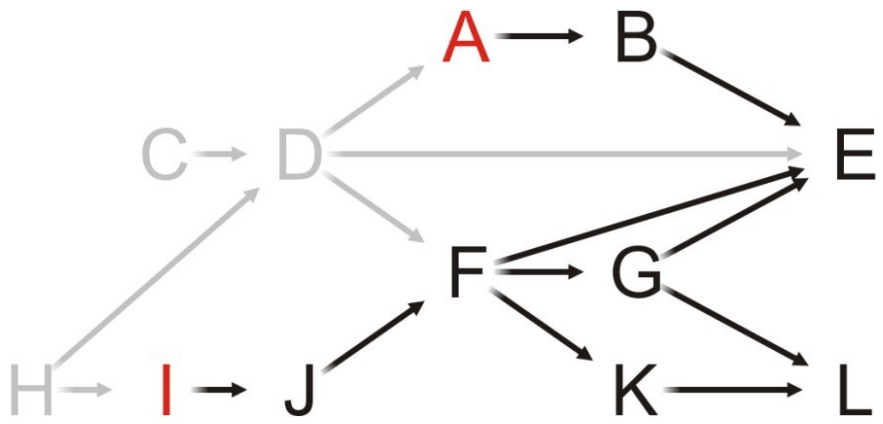
C, H

- We remove Task D, and now?



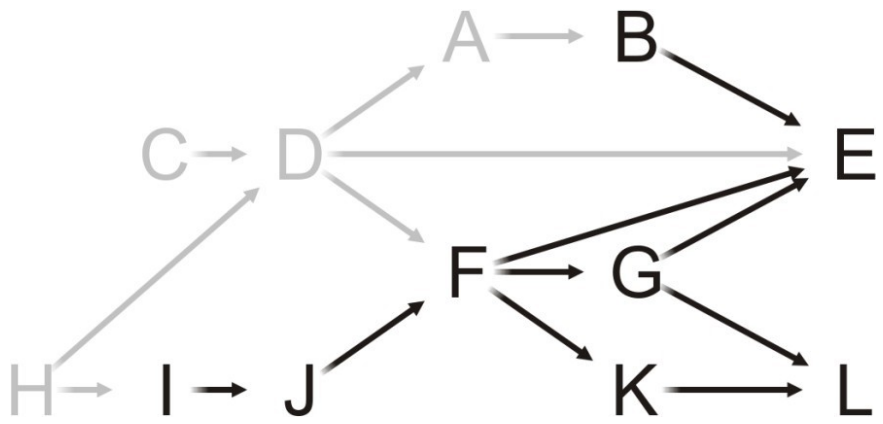
C, H, D

- Both Tasks A and I have in-degree zero
- Let’s choose Task A



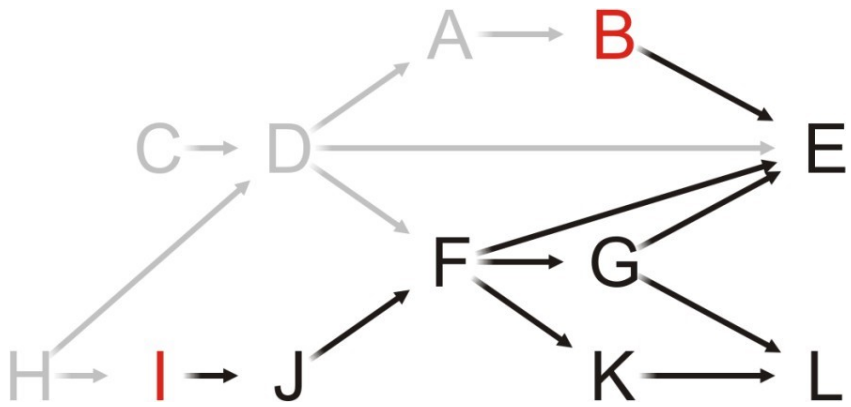
C, H, D

- Having removed A, what now?



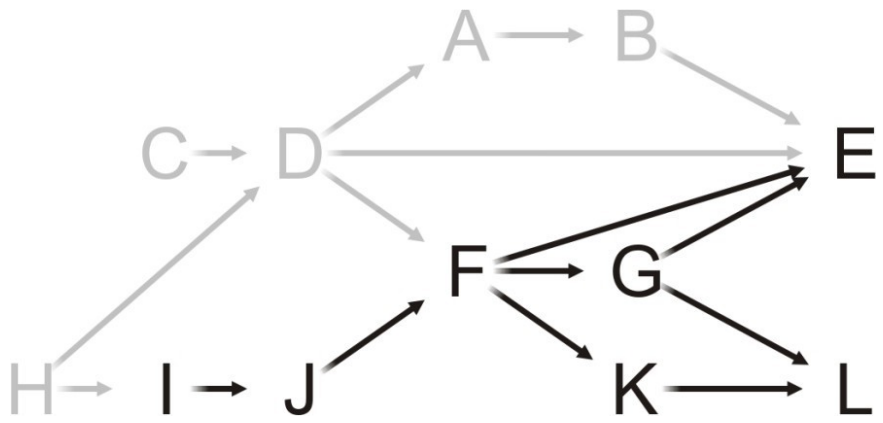
C, H, D, A

- Both Tasks B and I have in-degree zero
- Choose Task B



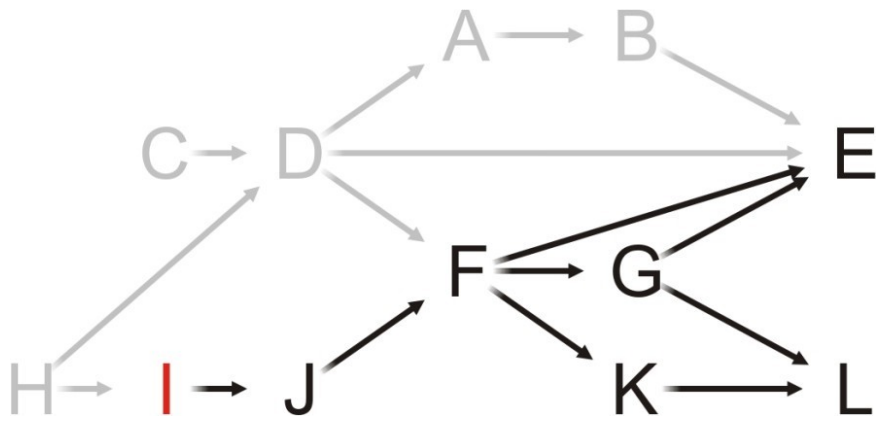
C, H, D, A

- Removing Task B, we note that Task E still has an in-degree of two
- Next?



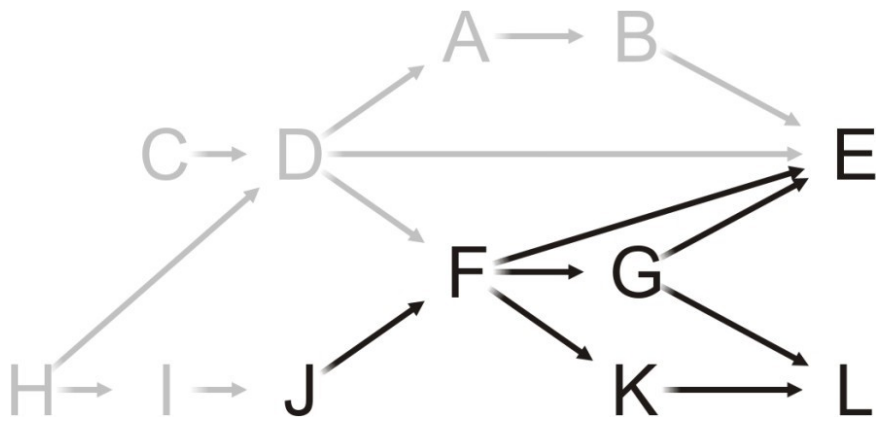
C, H, D, A, B

- As only Task I has in-degree zero, we choose it



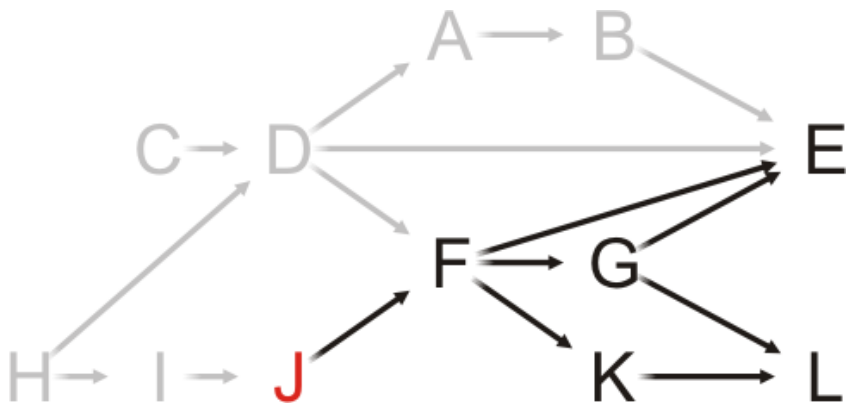
C, H, D, A, B

- Having completed Task I, what now?



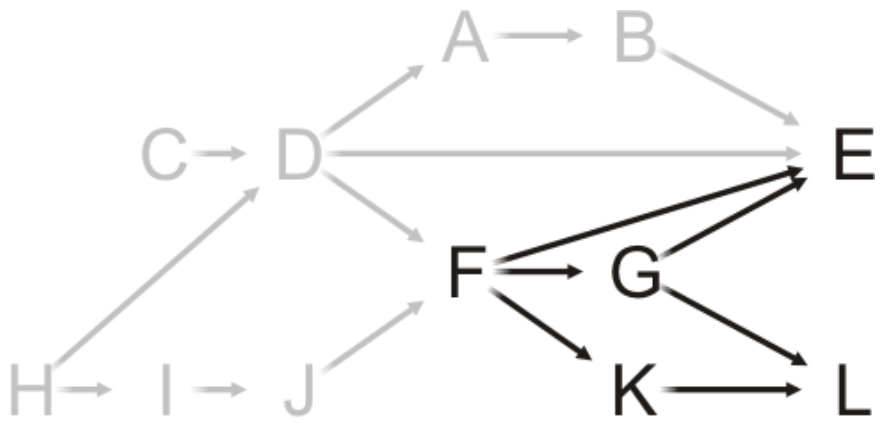
C, H, D, A, B, I

- Only Task J has in-degree zero: choose it



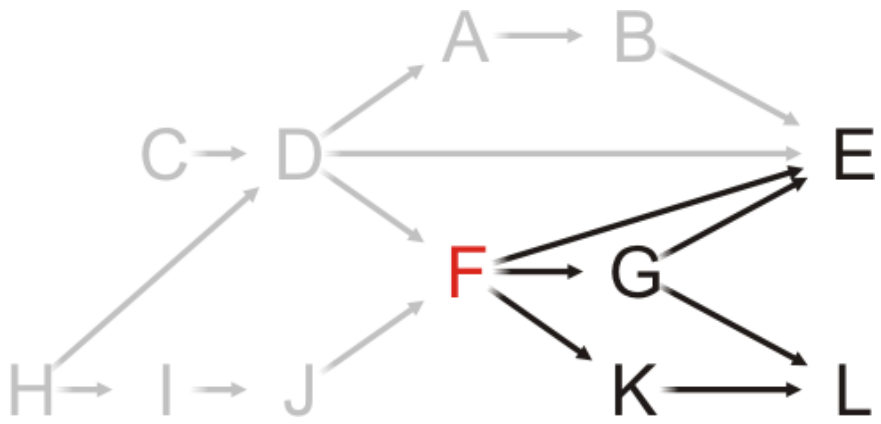
C, H, D, A, B, I

- Having completed Task J, what now?



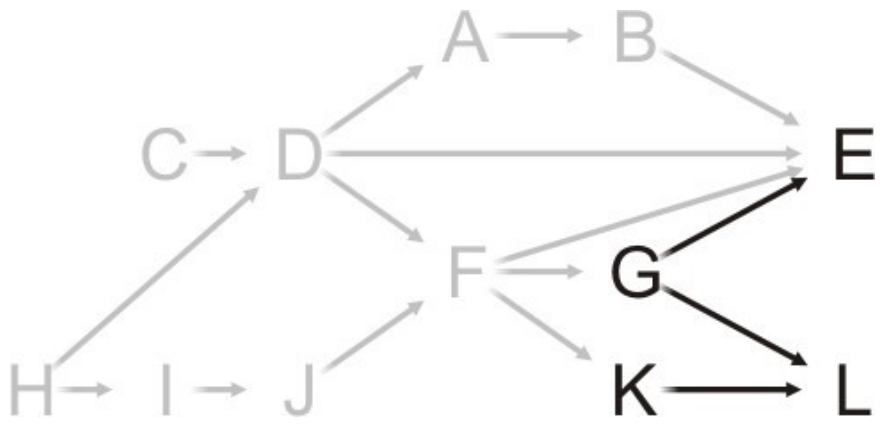
C, H, D, A, B, I, J

- Only Task F can be completed, so choose it



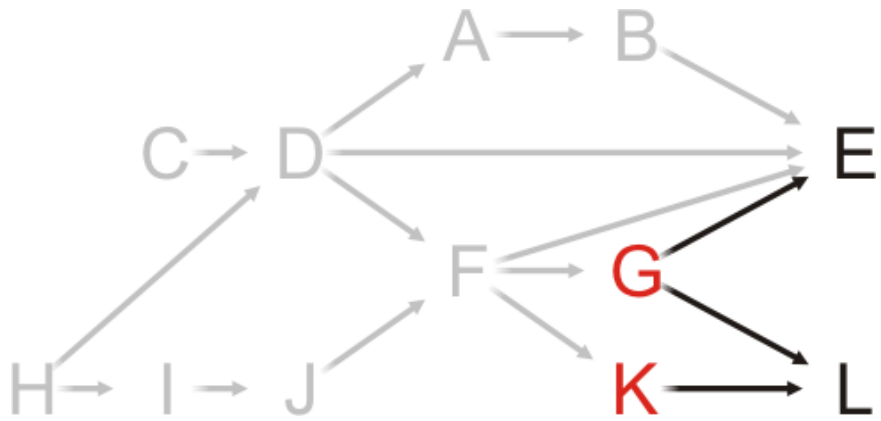
C, H, D, A, B, I, J

- What choices do we have now?



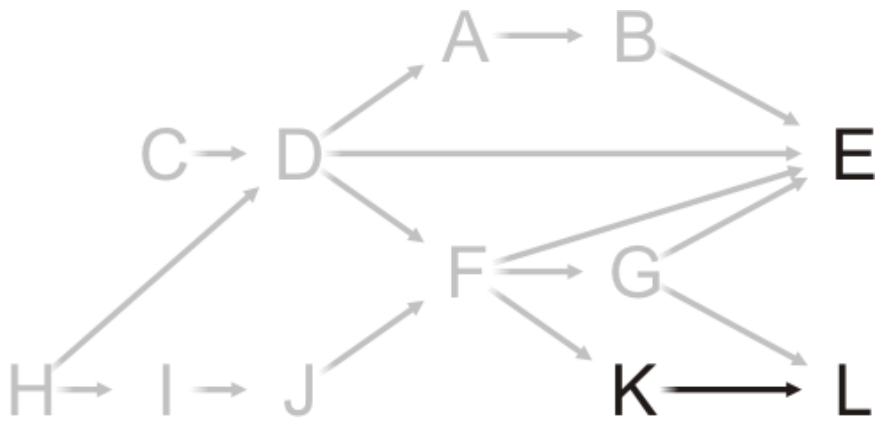
C, H, D, A, B, I, J, F

- We can perform Tasks G or K
- Choose Task G



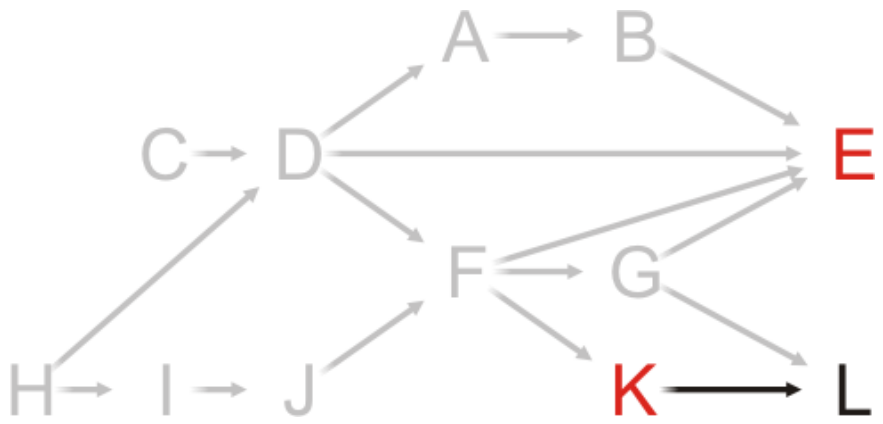
C, H, D, A, B, I, J, F

- Having removed Task G from the graph, what next?



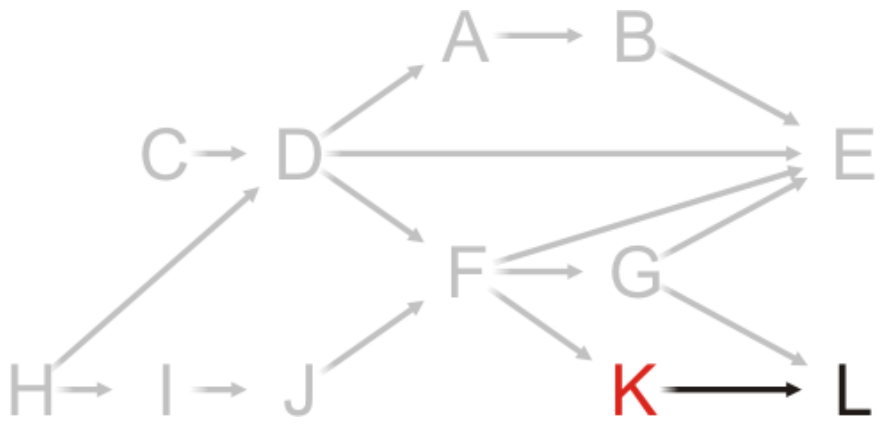
C, H, D, A, B, I, J, F, G

- Choosing between Tasks E and K, choose Task E



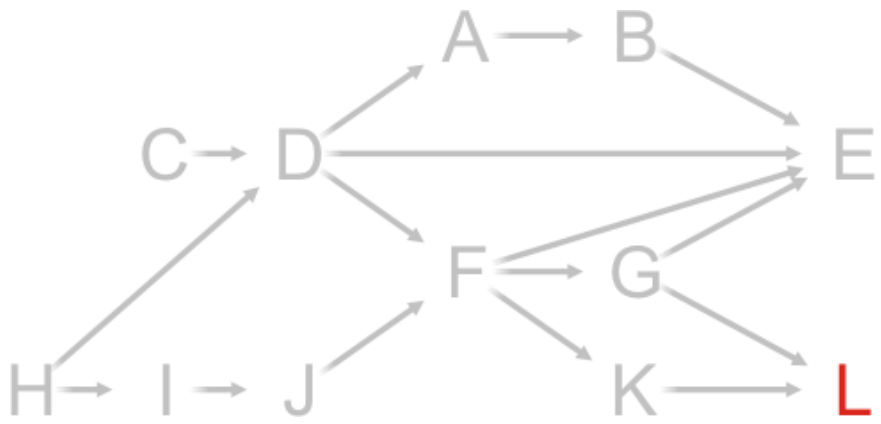
C, H, D, A, B, I, J, F, G

- At this point, Task K is the only one that can be run



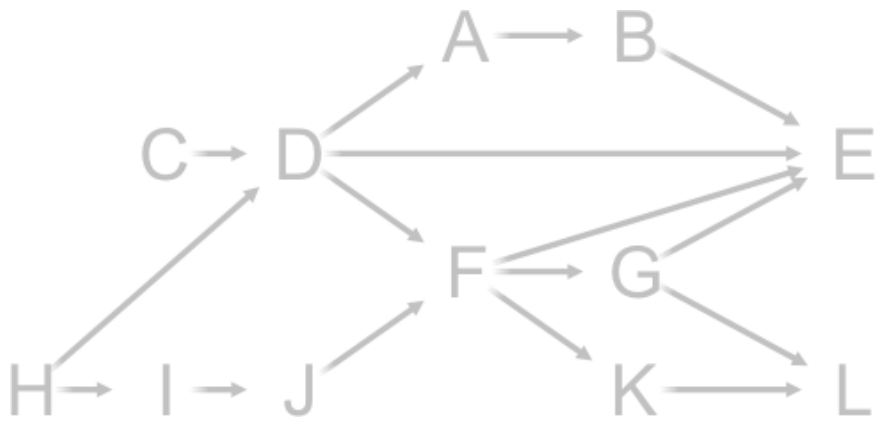
C, H, D, A, B, I, J, F, G, E

- And now that both Tasks G and K are complete, we can complete Task L



C, H, D, A, B, I, J, F, G, E, K

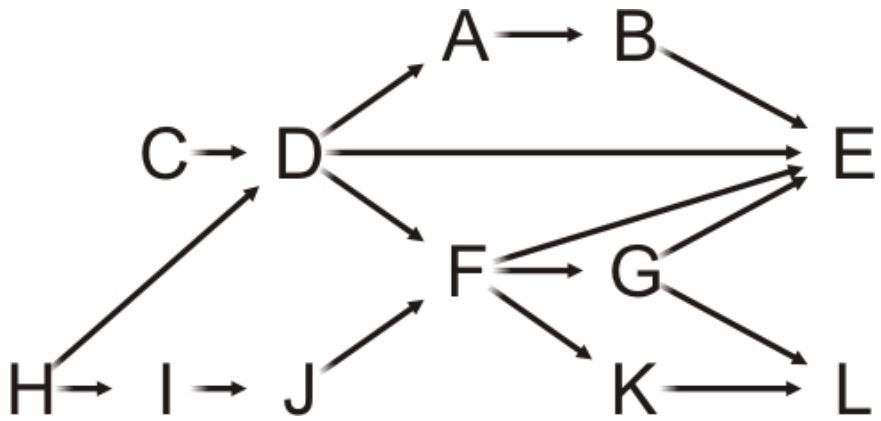
- There are no more vertices left



C, H, D, A, B, I, J, F, G, E, K, L

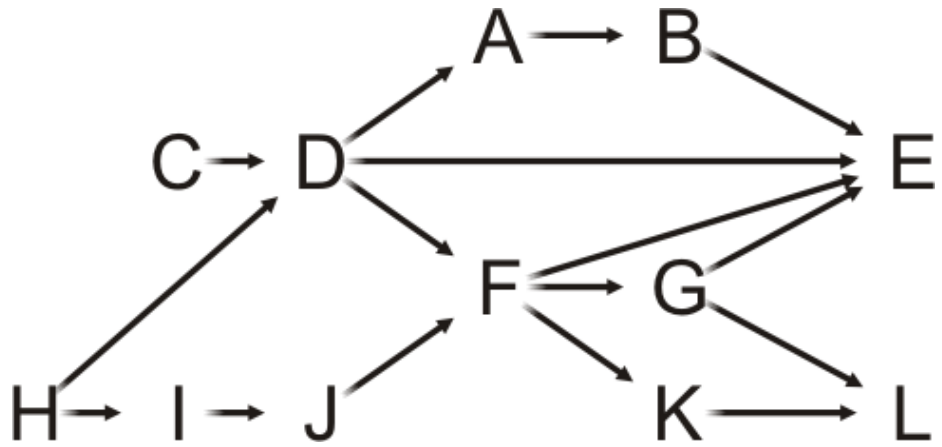
- Thus, one possible topological sort would be:

$$C, H, D, A, B, I, J, F, G, E, K, L$$



- Note that topological sorts need not be unique:

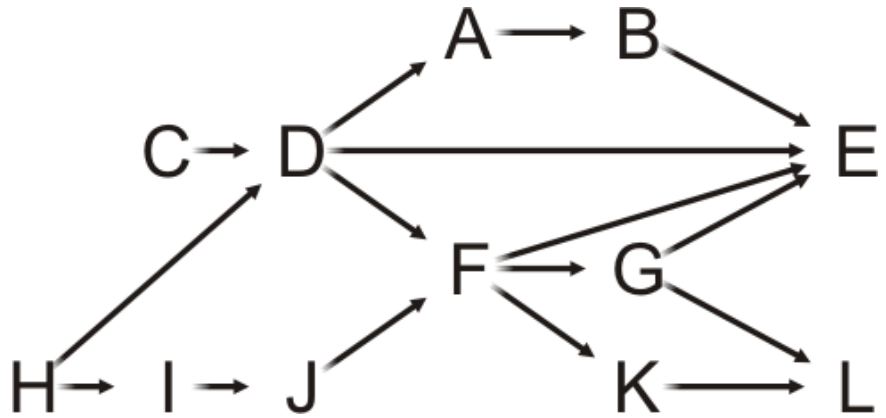
$$C, H, D, A, B, I, J, F, G, E, K, L$$
$$H, I, J, C, D, F, G, K, L, A, B, E$$



Analysis

- Firstly, find the **in-degrees** of each v.
  - Maintain a **table of the in-degrees** of each v.
  - Requires $O(?)$ memory.

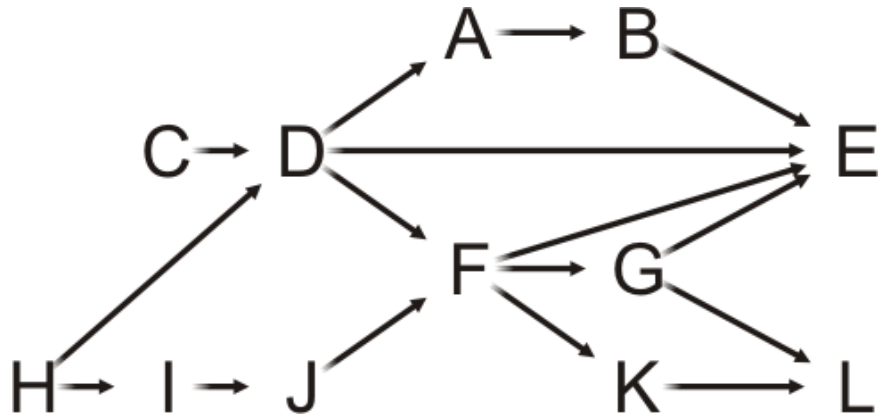
A	1
B	1
C	0
D	2
E	4
F	2
G	1
H	0
I	1
J	1
K	1
L	2



Analysis

- Firstly, find the **in-degrees** of each v.
  - Maintain a **table of the in-degrees** of each v.
  - Requires $O(V)$ memory.

A	1
B	1
C	0
D	2
E	4
F	2
G	1
H	0
I	1
J	1
K	1
L	2

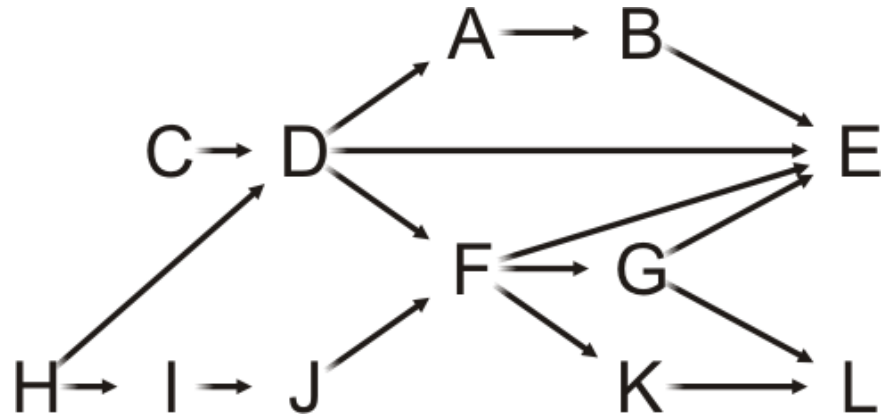


Analysis

- Firstly, find the **in-degrees** of each v.
  - Maintain a **table of the in-degrees** of each v.
  - Requires $O(V)$ memory.

- **Idea**: after visiting a vertex, find next vertex with in-degree $= 0$:
  - Overall Time complexity: $O(?)$

A	1
B	1
C	0
D	2
E	4
F	2
G	1
H	0
I	1
J	1
K	1
L	2

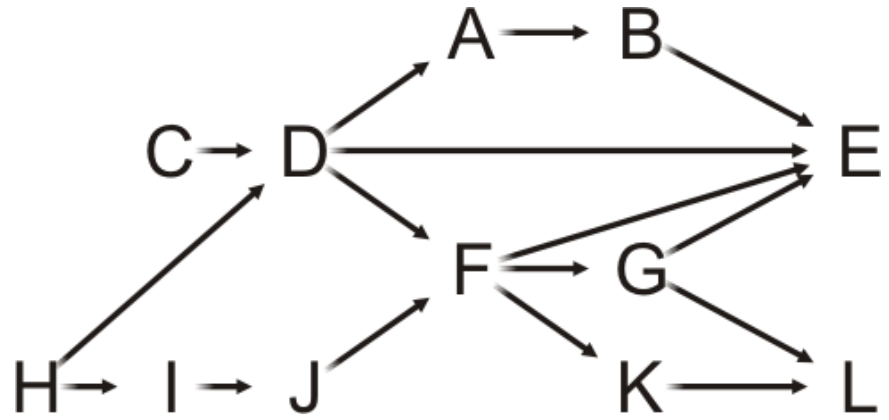


Analysis

- Firstly, find the **in-degrees** of each v.
  - Maintain a **table of the in-degrees** of each v.
  - Requires $O(V)$ memory.

- **Idea**: after visiting a vertex, find next vertex with in-degree $= 0$:
  - Overall Time complexity: $O(V^2)$

A	1
B	1
C	0
D	2
E	4
F	2
G	1
H	0
I	1
J	1
K	1
L	2



- Better idea:
  - Use **Queue** to temporarily store vertices with **in-degree zero**
  - Each time in-degree of a vertex is **decremented to zero**, Enqueue it.

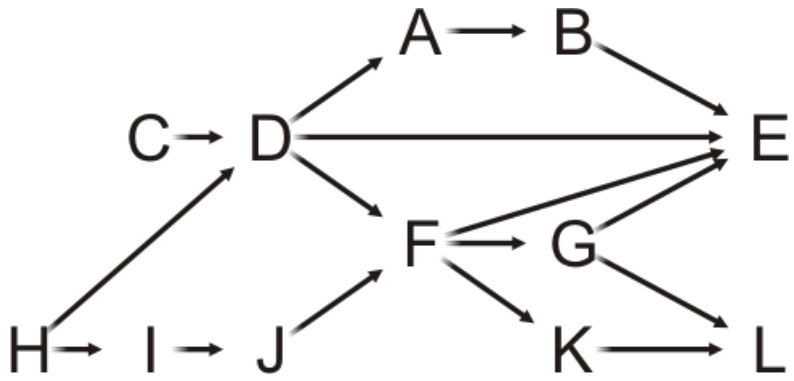
Topological Sort with Queue (Q)

- Initialization:
  - **Allocate memory** and **initialize** an array of in-degrees
  - Initialize Q with **all v having in-degree 0**

- **While** Q is **not empty**:
  - **Pop** a vertex from Q
  - **Decrement** the in-degree of each neighbor
  - Neighbors whose in-degree **becomes zero are pushed into Q**

Example

Stepping through the table, push all source vertices into the queue



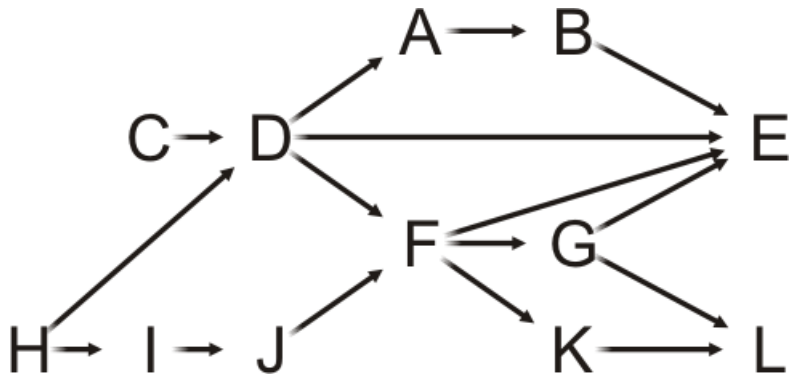
A	1
B	1
C	0
D	2
E	4
F	2
G	1
H	0
I	1
J	1
K	1
L	2

Queue:

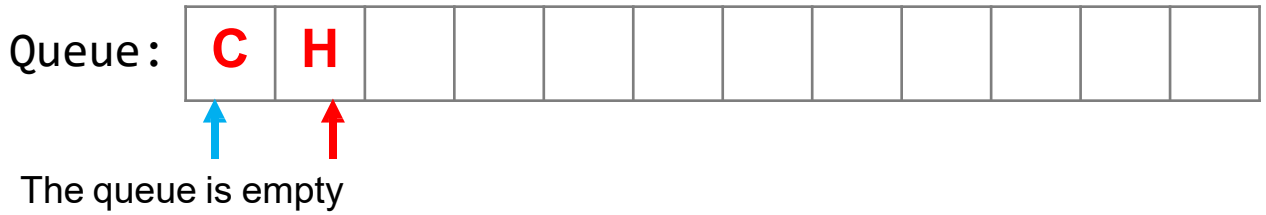
The queue is empty

Example

Stepping through the table, push all source vertices into the queue

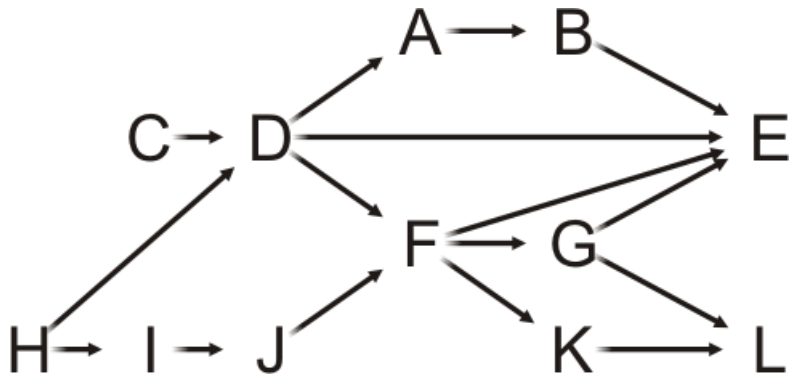


A	1
B	1
C	**0**
D	2
E	4
F	2
G	1
H	**0**
I	1
J	1
K	1
L	2



Example

Pop the front of the queue



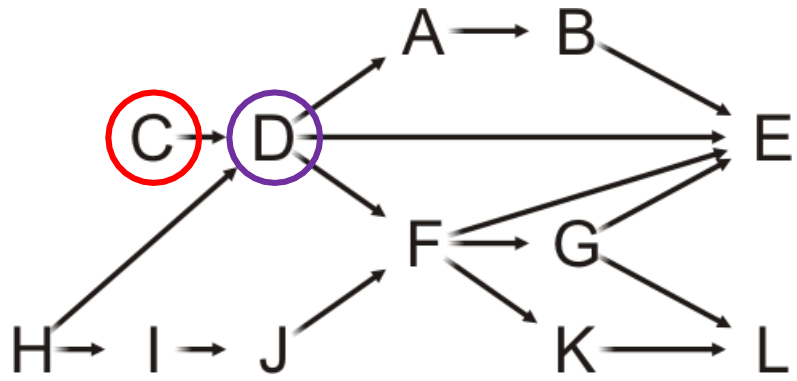
Queue: C H

A	1
B	1
C	0
D	2
E	4
F	2
G	1
H	0
I	1
J	1
K	1
L	2

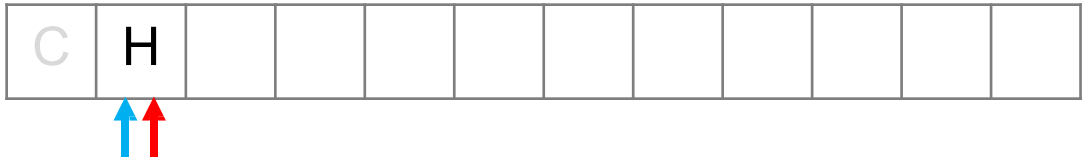
Example

Pop the front of the queue

- C has one neighbor: D



Queue:

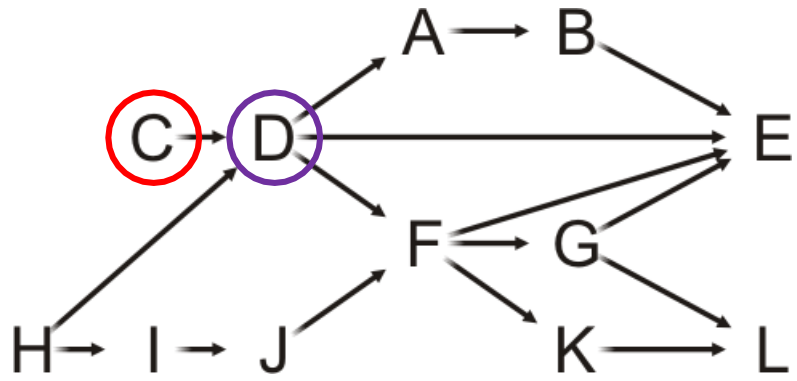


A	1
B	1
C	**0**
D	**2**
E	4
F	2
G	1
H	0
I	1
J	1
K	1
L	2

Example

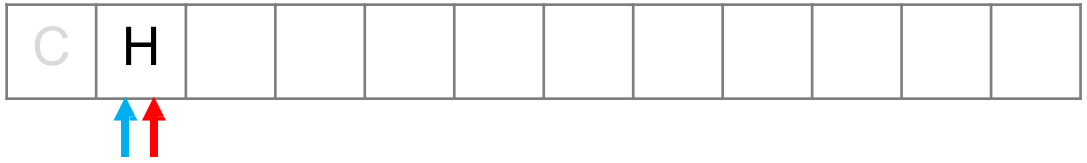
Pop the front of the queue

- C has one neighbor: D
- Decrement its in-degree



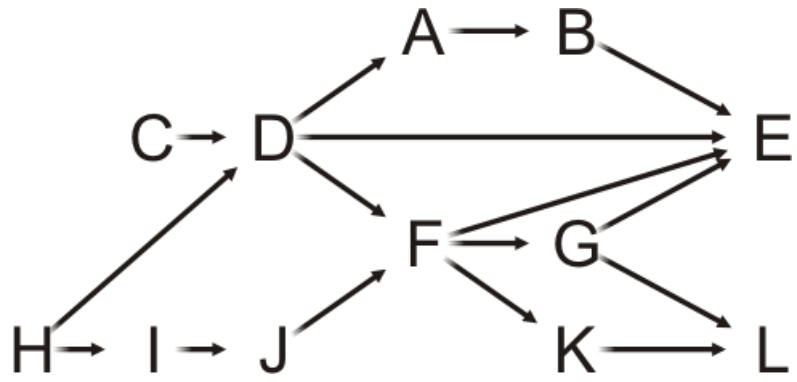
A	1
B	1
C	0
D	1
E	4
F	2
G	1
H	0
I	1
J	1
K	1
L	2

Queue:

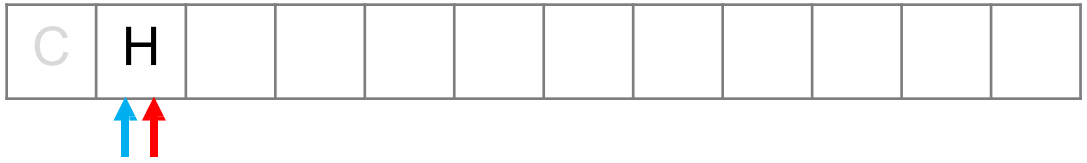


Example

Pop the front of the queue



Queue:

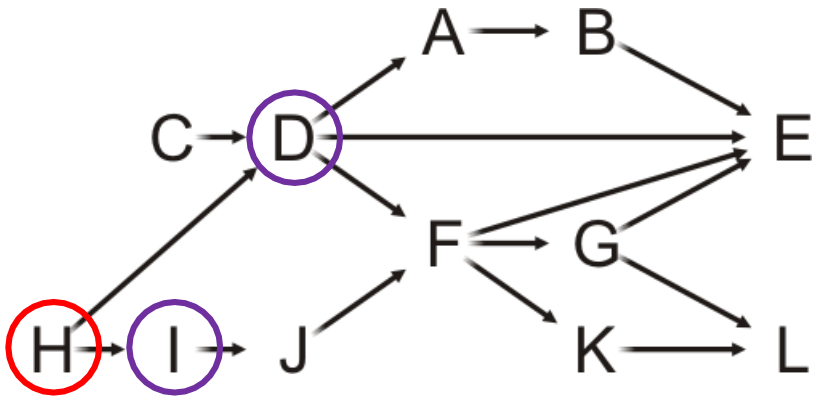


A	1
B	1
C	0
D	1
E	4
F	2
G	1
H	0
I	1
J	1
K	1
L	2

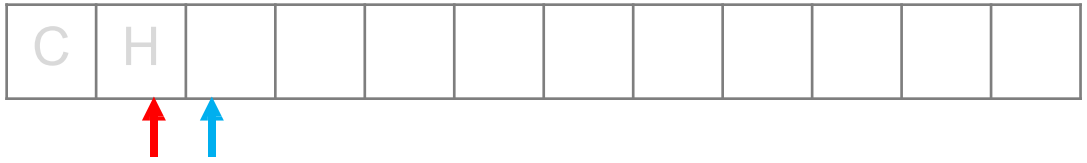
Example

Pop the front of the queue

- H has two neighbors: D and I



Queue:

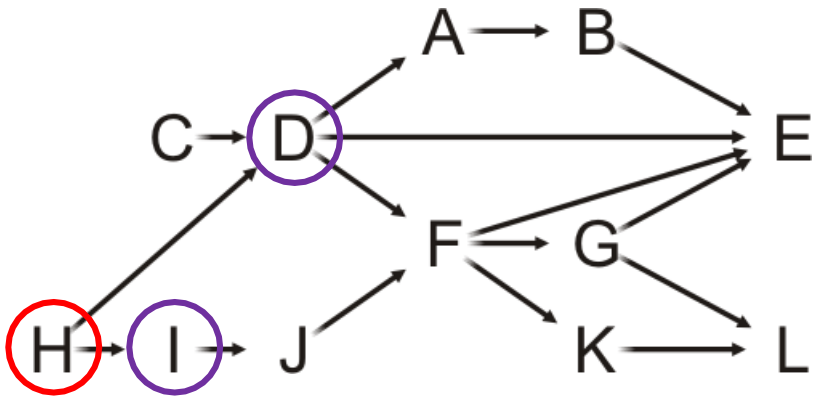


A	1
B	1
C	0
D	**1**
E	4
F	2
G	1
H	**0**
I	**1**
J	1
K	1
L	2

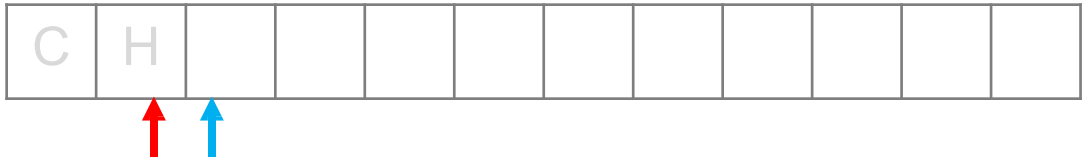
Example

Pop the front of the queue

- H has two neighbors: D and I
- Decrement their in-degrees



Queue:

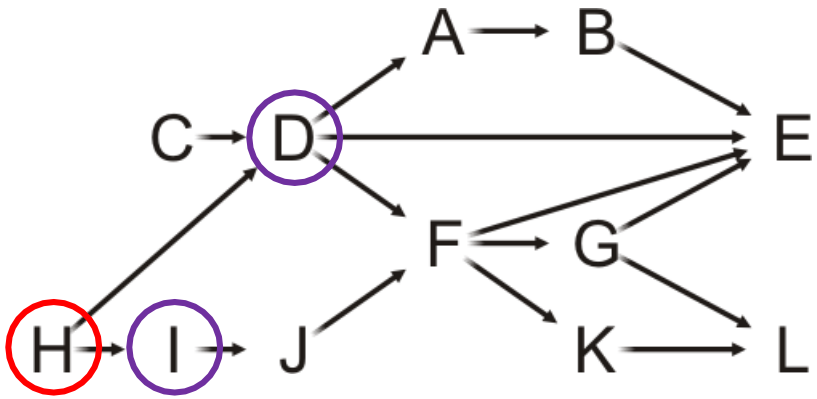


A	1
B	1
C	0
D	**0**
E	4
F	2
G	1
H	**0**
I	**0**
J	1
K	1
L	2

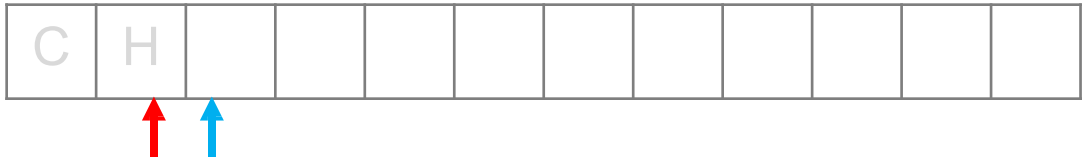
Example

Pop the front of the queue

- H has two neighbors: D and I
- Decrement their in-degrees
  - Both are decremented to zero, so push them onto the queue



Queue:

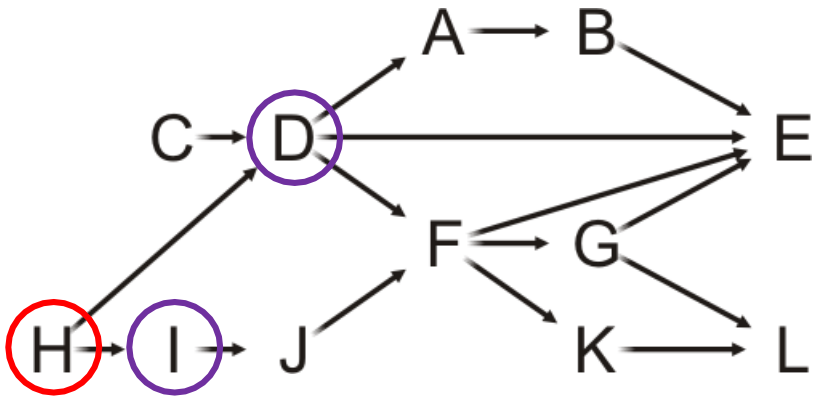


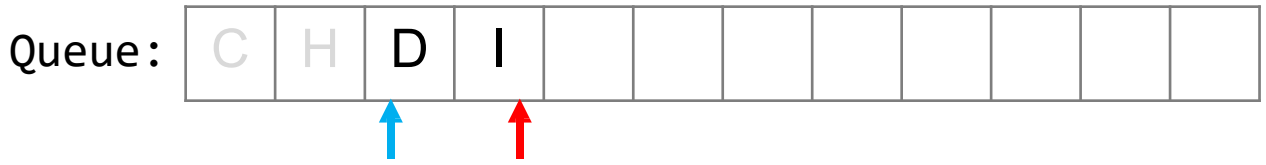
A	1
B	1
C	0
D	**0**
E	4
F	2
G	1
H	**0**
I	**0**
J	1
K	1
L	2

Example

Pop the front of the queue

- H has two neighbors: D and I
- Decrement their in-degrees
  - Both are decremented to zero, so push them onto the queue

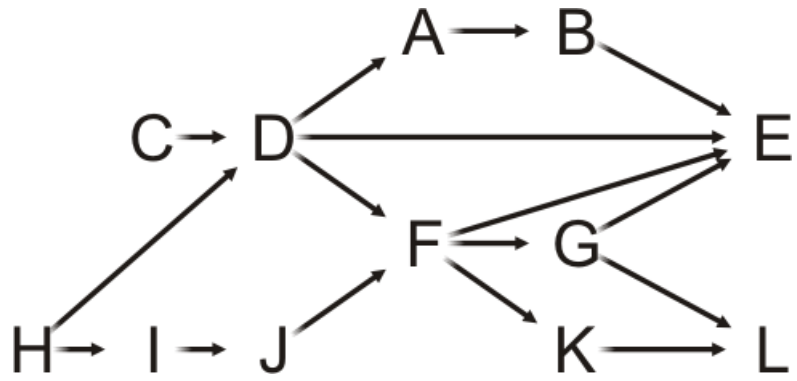


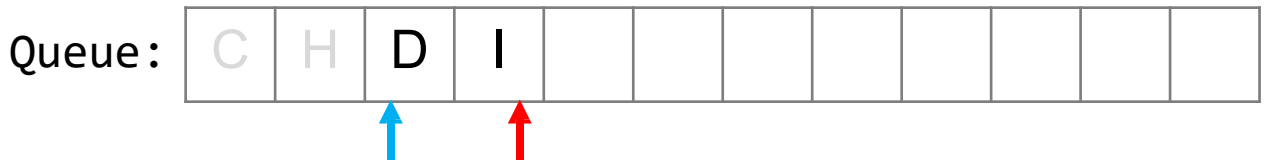


A	1
B	1
C	0
D	**0**
E	4
F	2
G	1
H	**0**
I	**0**
J	1
K	1
L	2

Example

Pop the front of the queue



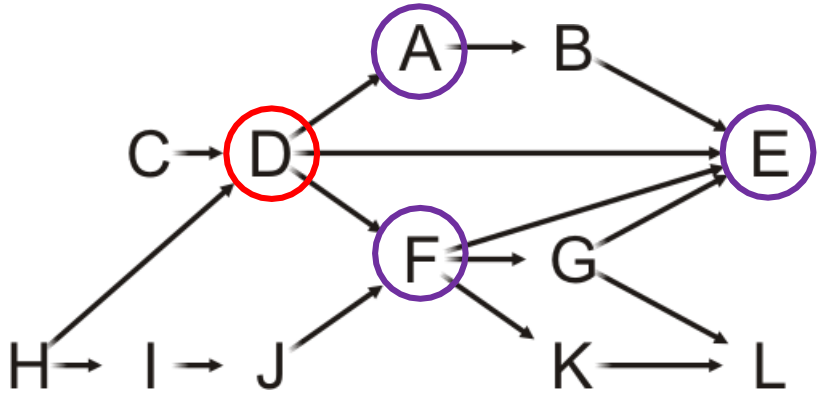


A	1
B	1
C	0
D	0
E	4
F	2
G	1
H	0
I	0
J	1
K	1
L	2

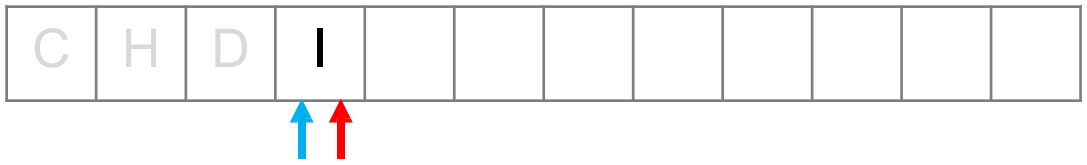
Example

Pop the front of the queue

- D has three neighbors: A, E and F



Queue:

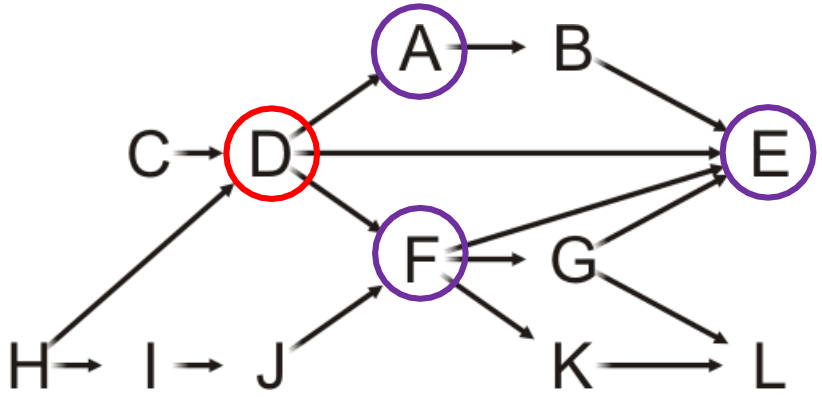


A	1
B	1
C	0
D	0
E	4
F	2
G	1
H	0
I	0
J	1
K	1
L	2

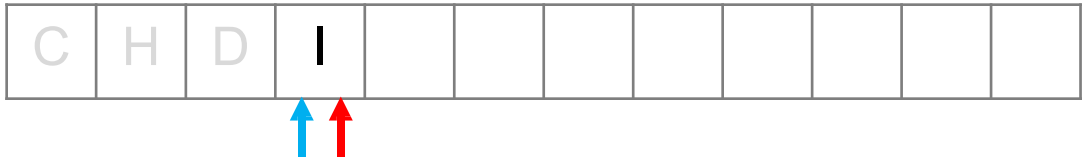
Example

Pop the front of the queue

- D has three neighbors: A, E and F
- Decrement their in-degrees



Queue:

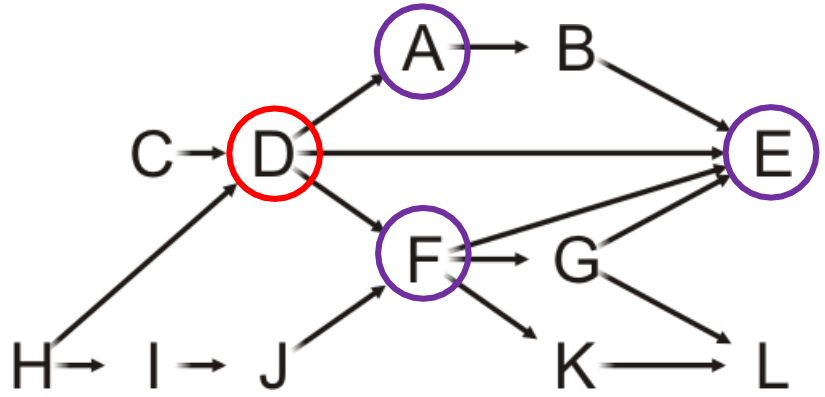


A	0
B	1
C	0
D	0
E	3
F	1
G	1
H	0
I	0
J	1
K	1
L	2

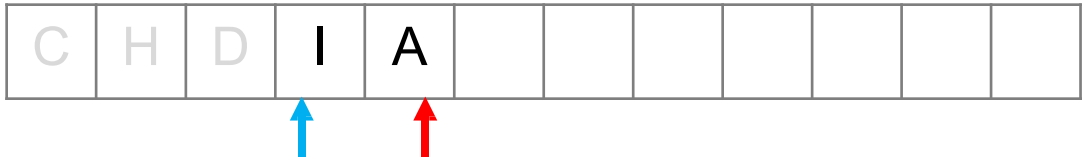
Example

Pop the front of the queue

- D has three neighbors: A, E and F
- Decrement their in-degrees
  - A is decremented to zero, so push it onto the queue



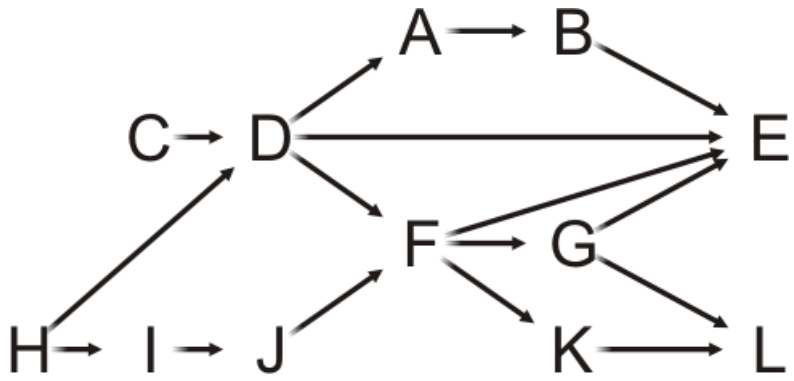
Queue:



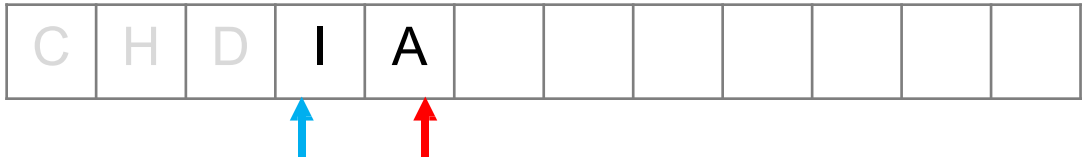
A	0
B	1
C	0
D	0
E	3
F	1
G	1
H	0
I	0
J	1
K	1
L	2

Example

Pop the front of the queue



Queue:

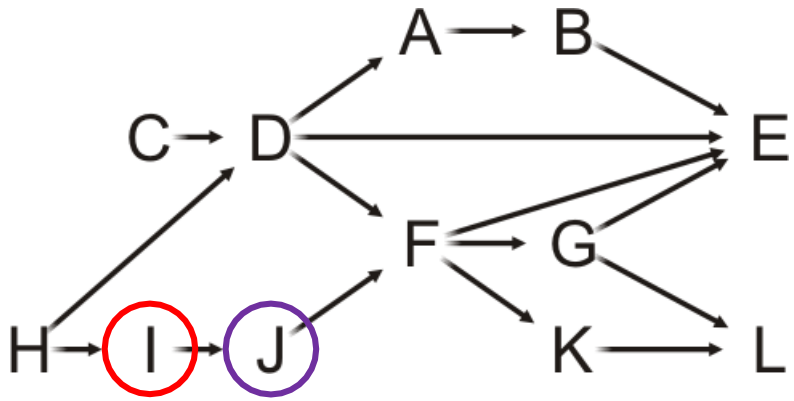


A	0
B	1
C	0
D	0
E	3
F	1
G	1
H	0
I	0
J	1
K	1
L	2

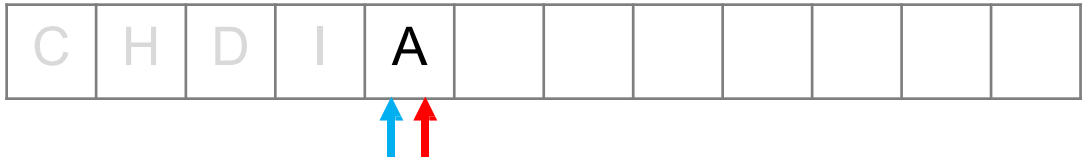
Example

Pop the front of the queue

- I has one neighbor: J



Queue:

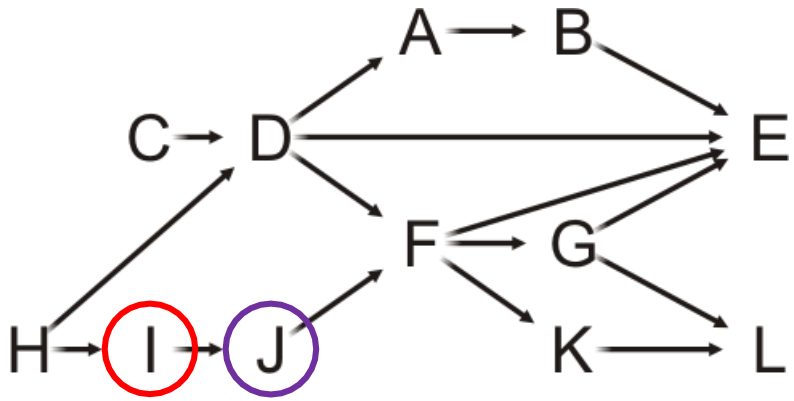


A	0
B	1
C	0
D	0
E	3
F	1
G	1
H	0
I	**0**
J	**1**
K	1
L	2

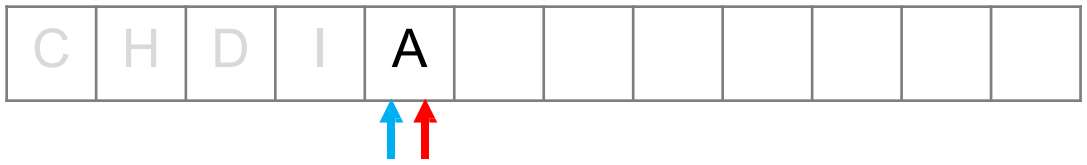
Example

Pop the front of the queue

- I has one neighbor: J
- Decrement its in-degree



Queue:

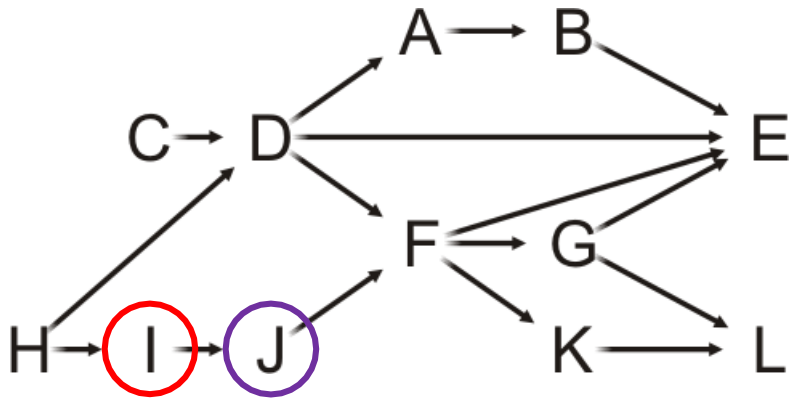


A	0
B	1
C	0
D	0
E	3
F	1
G	1
H	0
I	**0**
J	**0**
K	1
L	2

Example

Pop the front of the queue

- I has one neighbor:J
- Decrement its in-degree
  - J is decremented to zero, so push it onto the queue



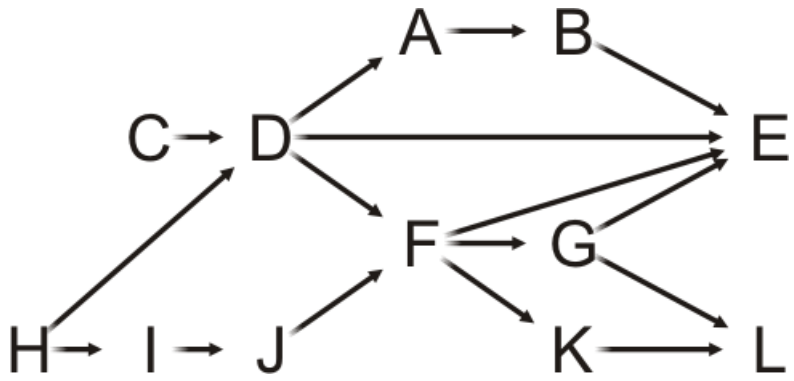
Queue:

C	H	D	I	A	J						

A	0
B	1
C	0
D	0
E	3
F	1
G	1
H	0
I	**0**
J	**0**
K	1
L	2

Example

Pop the front of the queue



Queue:

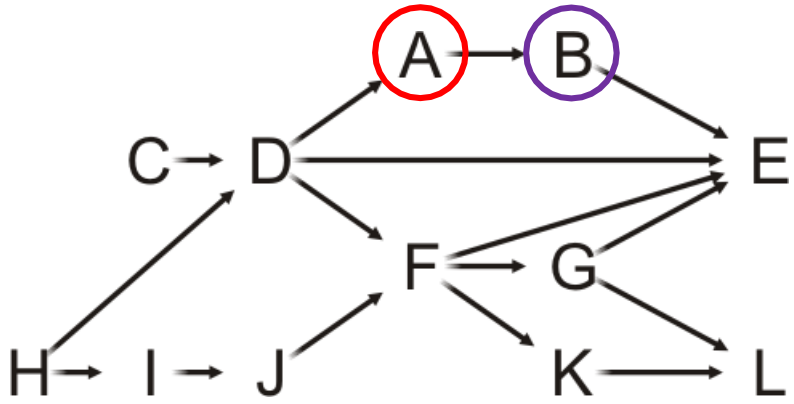
C	H	D	I	A	J						

A	0
B	1
C	0
D	0
E	3
F	1
G	1
H	0
I	0
J	0
K	1
L	2

Example

Pop the front of the queue

- A has one neighbor: B



Queue:

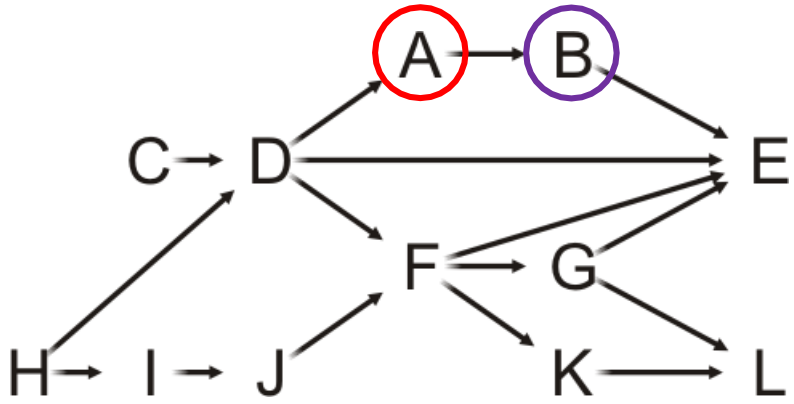
C	H	D	I	A	J						

A	0
B	1
C	0
D	0
E	3
F	1
G	1
H	0
I	0
J	0
K	1
L	2

Example

Pop the front of the queue

- A has one neighbor: B
- Decrement its in-degree



Queue:

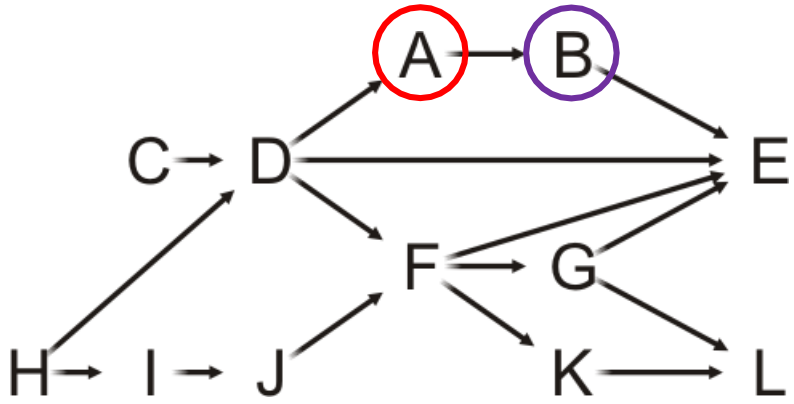
C	H	D	I	A	J						

A	0
B	0
C	0
D	0
E	3
F	1
G	1
H	0
I	0
J	0
K	1
L	2

Example

Pop the front of the queue

- A has one neighbor: B
- Decrement its in-degree
  - B is decremented to zero, so push it onto the queue



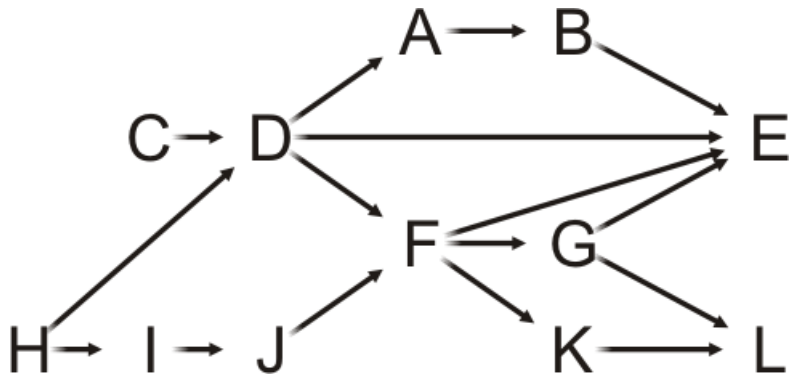
Queue:

C	H	D	I	A	J	B					

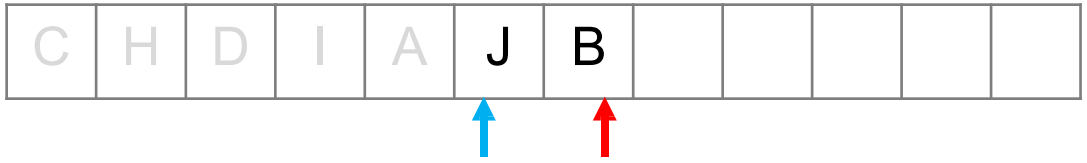
A	0
B	0
C	0
D	0
E	3
F	1
G	1
H	0
I	0
J	0
K	1
L	2

Example

Pop the front of the queue



Queue:

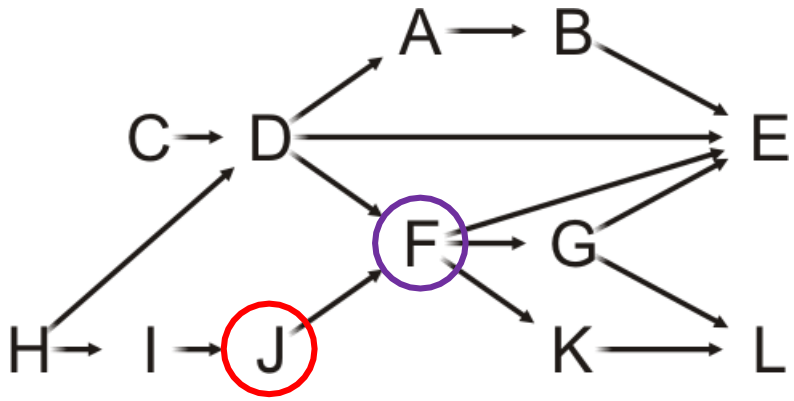


A	0
B	0
C	0
D	0
E	3
F	1
G	1
H	0
I	0
J	0
K	1
L	2

Example

Pop the front of the queue

- J has one neighbor: F



Queue:

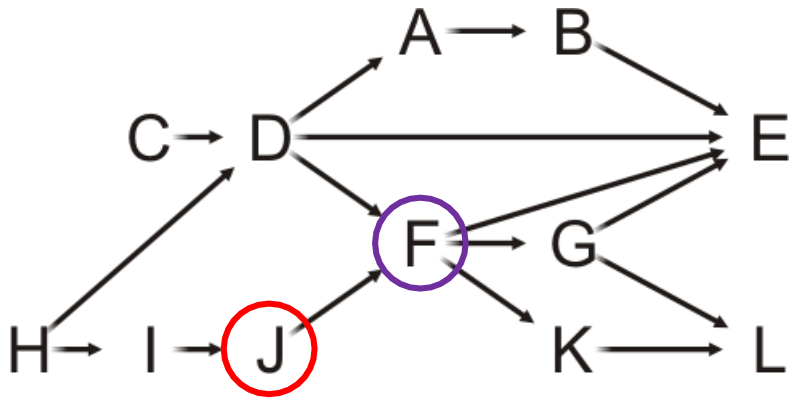
C	H	D	I	A	J	B					

A	0
B	0
C	0
D	0
E	3
F	**1**
G	1
H	0
I	0
J	**0**
K	1
L	2

Example

Pop the front of the queue

- J has one neighbor: F
- Decrement its in-degree



Queue:

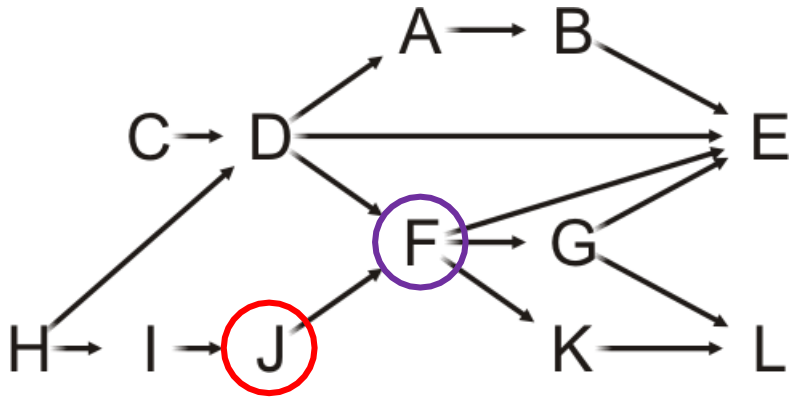
C	H	D	I	A	J	B					

A	0
B	0
C	0
D	0
E	3
F	**0**
G	1
H	0
I	0
J	**0**
K	1
L	2

Example

Pop the front of the queue

- J has one neighbor: F
- Decrement its in-degree
  - F is decremented to zero, so push it onto the queue



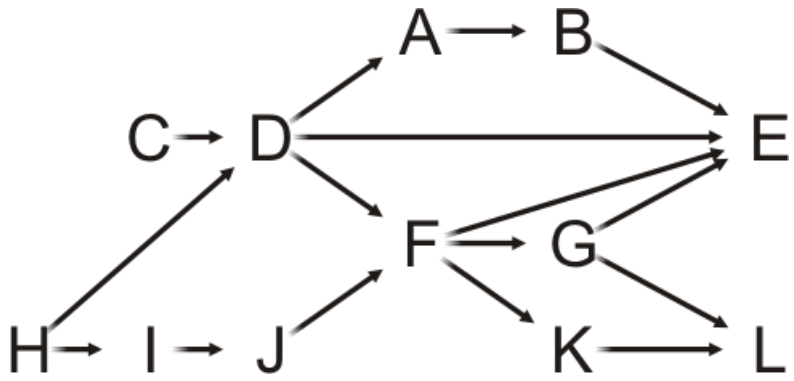
Queue:

C	H	D	I	A	J	B	F				

A	0
B	0
C	0
D	0
E	3
F	**0**
G	1
H	0
I	0
J	**0**
K	1
L	2

Example

Pop the front of the queue



Queue:

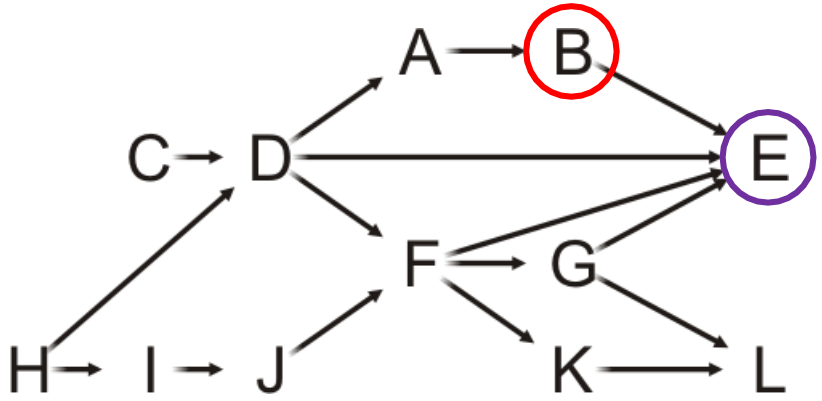
C	H	D	I	A	J	B	F				

A	0
B	0
C	0
D	0
E	3
F	0
G	1
H	0
I	0
J	0
K	1
L	2

Example

Pop the front of the queue

- B has one neighbor: E



Queue:

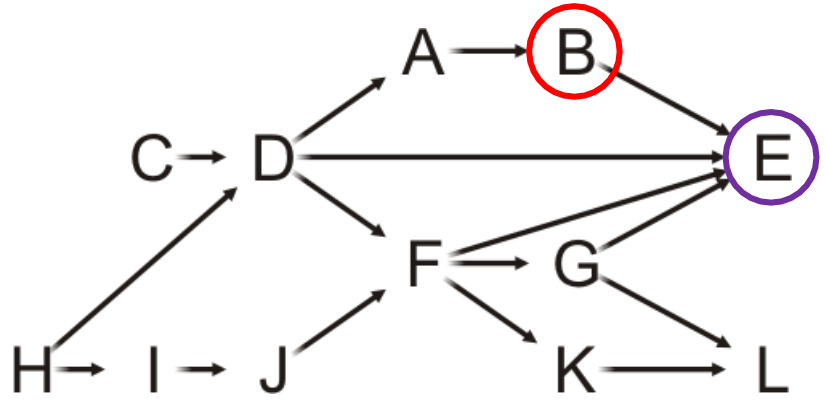
C	H	D	I	A	J	B	F				

A	0
B	**0**
C	0
D	0
E	**3**
F	0
G	1
H	0
I	0
J	0
K	1
L	2

Example

Pop the front of the queue

- B has one neighbor: E
- Decrement its in-degree



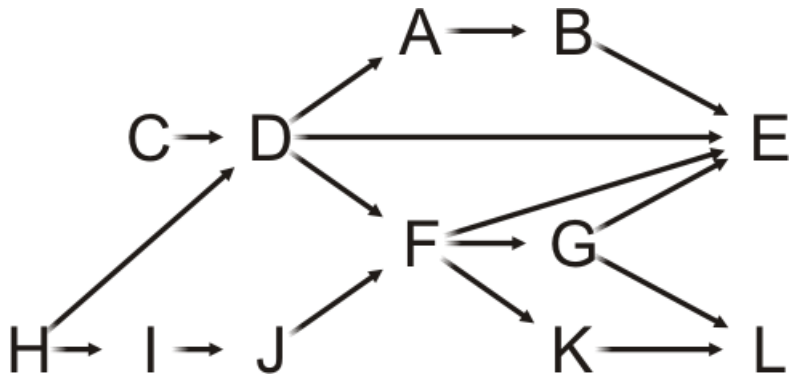
Queue:

C	H	D	I	A	J	B	F				

A	0
B	**0**
C	0
D	0
E	**2**
F	0
G	1
H	0
I	0
J	0
K	1
L	2

Example

Pop the front of the queue



Queue:

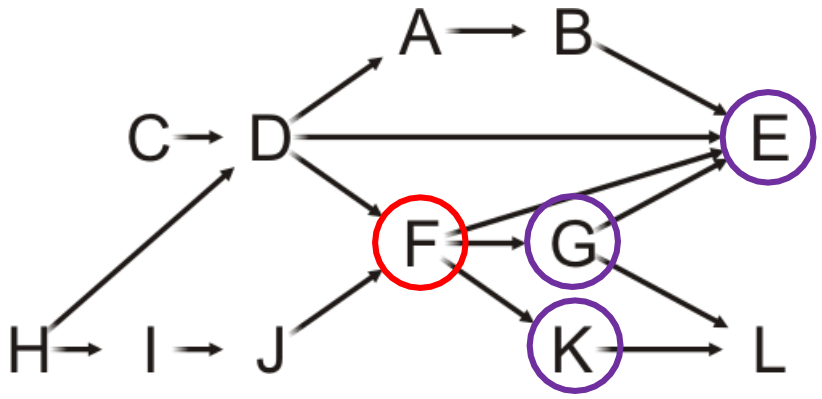
C	H	D	I	A	J	B	F				

A	0
B	0
C	0
D	0
E	2
F	0
G	1
H	0
I	0
J	0
K	1
L	2

Example

Pop the front of the queue

- F has three neighbors: E, G and K



Queue:

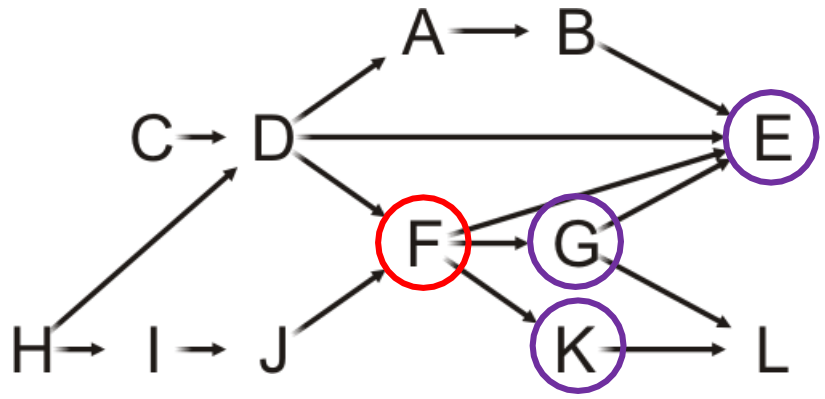
C	H	D	I	A	J	B	F				

A	0
B	0
C	0
D	0
E	**2**
F	**0**
G	**1**
H	0
I	0
J	0
K	**1**
L	2

Example

Pop the front of the queue

- F has three neighbors: E, G and K
- Decrement their in-degrees



Queue:

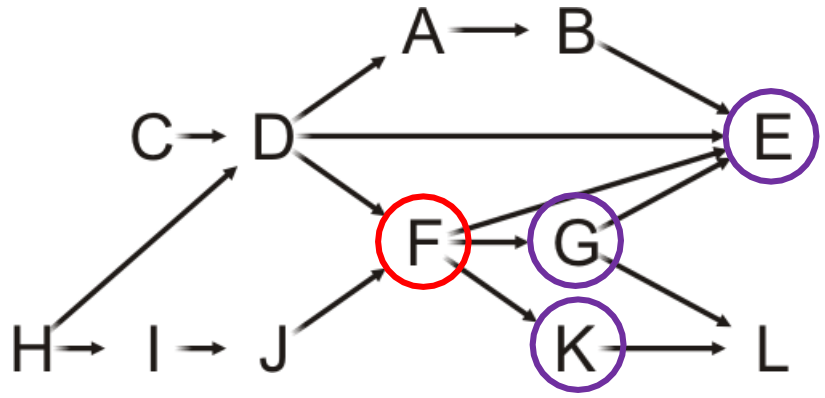
C	H	D	I	A	J	B	F				

A	0
B	0
C	0
D	0
E	**1**
F	**0**
G	**0**
H	0
I	0
J	0
K	**0**
L	2

Example

Pop the front of the queue

- F has three neighbors: E, G and K
- Decrement their in-degrees
  - G and K are decremented to zero, so push them onto the queue



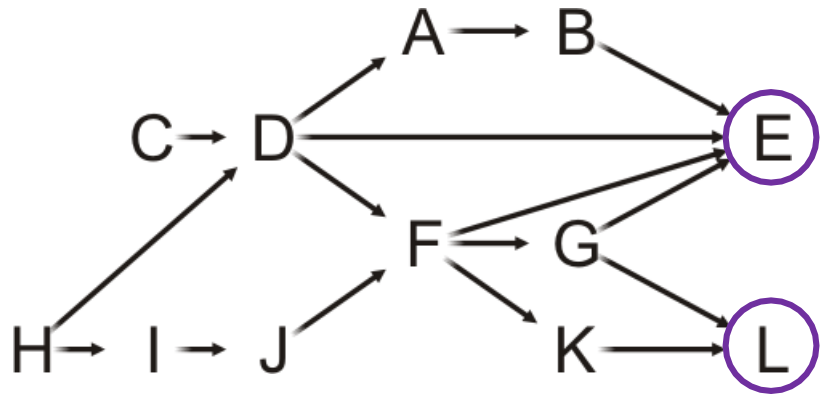
Queue:

C	H	D	I	A	J	B	F	G	K		

A	0
B	0
C	0
D	0
E	**1**
F	**0**
G	**0**
H	0
I	0
J	0
K	**0**
L	2

Example

Pop the front of the queue



Queue:

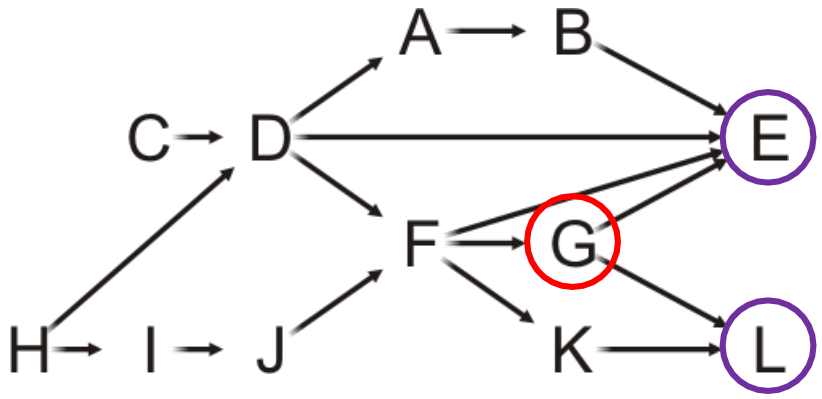
C	H	D	I	A	J	B	F	G	K		

A	0
B	0
C	0
D	0
E	1
F	0
G	0
H	0
I	0
J	0
K	0
L	2

Example

Pop the front of the queue

- G has two neighbors: E and L



Queue:

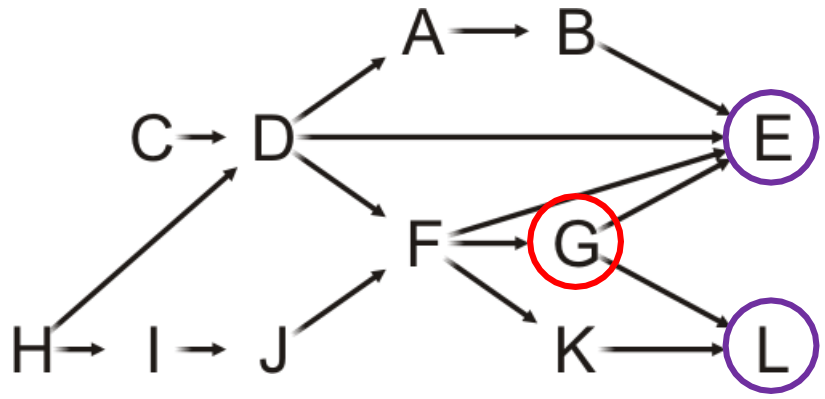
C	H	D	I	A	J	B	F	G	K		

A	0
B	0
C	0
D	0
E	**1**
F	0
G	**0**
H	0
I	0
J	0
K	0
L	**2**

Example

Pop the front of the queue

- G has two neighbors: E and L
- Decrement their in-degrees



Queue:

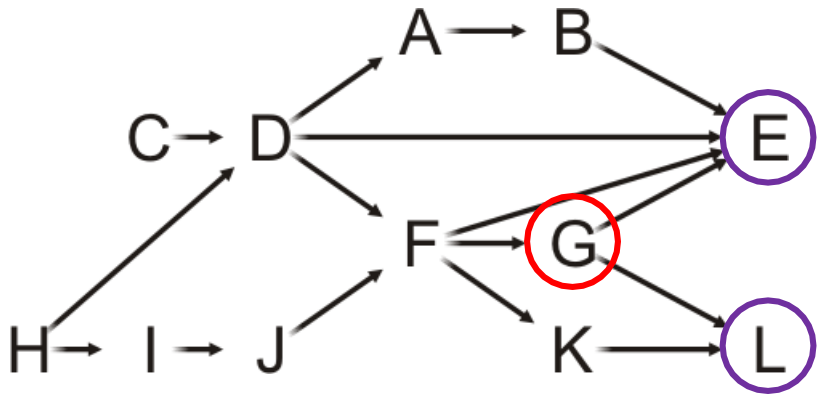
C	H	D	I	A	J	B	F	G	K		

A	0
B	0
C	0
D	0
E	**0**
F	0
G	**0**
H	0
I	0
J	0
K	0
L	**1**

Example

Pop the front of the queue

- G has two neighbors: E and L
- Decrement their in-degrees
  - E is decremented to zero, so push it onto the queue



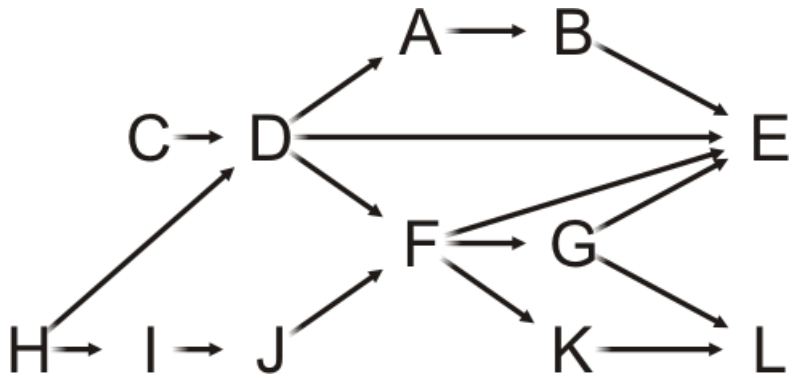
Queue:

C	H	D	I	A	J	B	F	G	K	E	

A	0
B	0
C	0
D	0
E	**0**
F	0
G	**0**
H	0
I	0
J	0
K	0
L	**1**

Example

Pop the front of the queue



Queue:

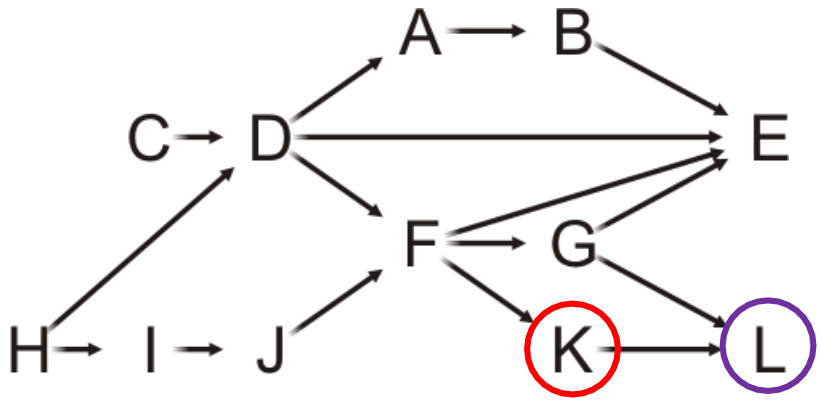
C	H	D	I	A	J	B	F	G	K	E	

A	0
B	0
C	0
D	0
E	0
F	0
G	0
H	0
I	0
J	0
K	0
L	1

Example

Pop the front of the queue

- K has one neighbors: L



Queue:

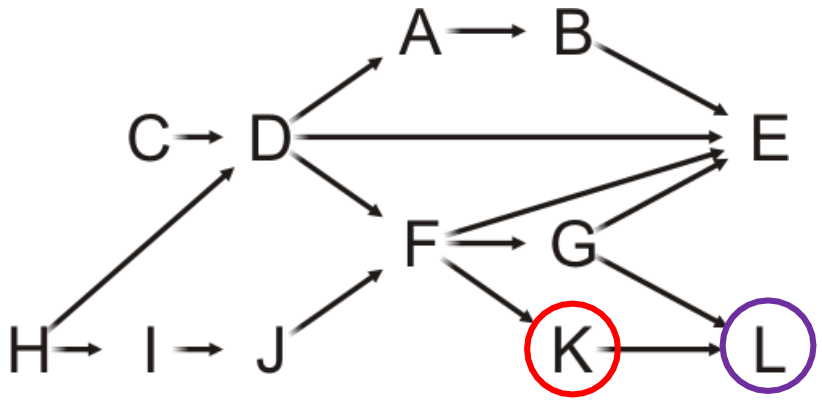
C	H	D	I	A	J	B	F	G	K	E	

A	0
B	0
C	0
D	0
E	0
F	0
G	0
H	0
I	0
J	0
K	**0**
L	**1**

Example

Pop the front of the queue

- K has one neighbors: L
- Decrement its in-degree



Queue:

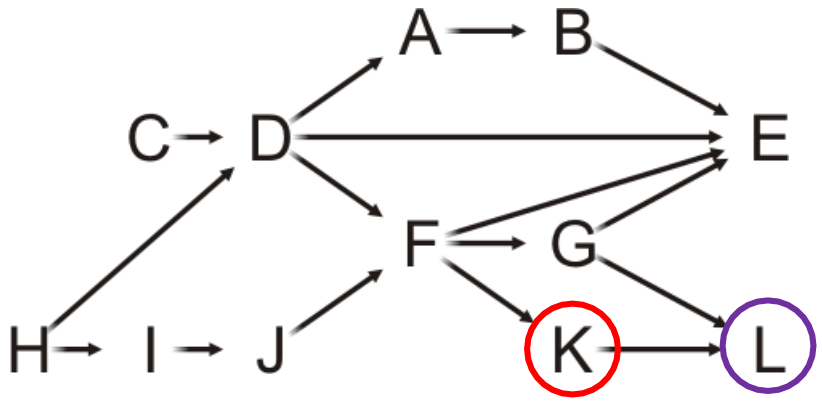
C	H	D	I	A	J	B	F	G	K	E	

A	0
B	0
C	0
D	0
E	0
F	0
G	0
H	0
I	0
J	0
K	**0**
L	**0**

Example

Pop the front of the queue

- K has one neighbors: L
- Decrement its in-degree
  - L is decremented to zero, so push it onto the queue



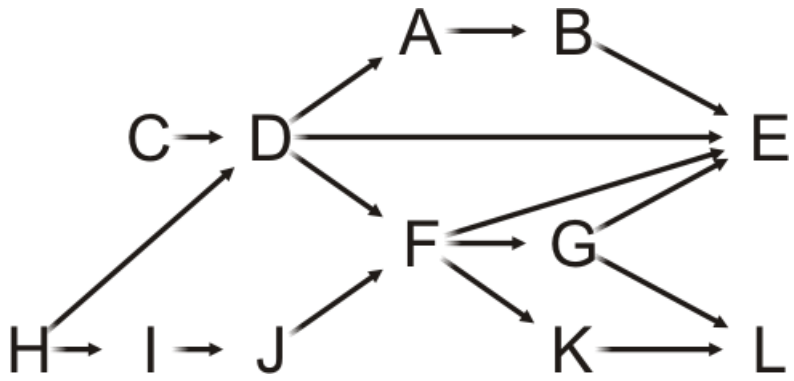
Queue:

C	H	D	I	A	J	B	F	G	K	E	L

A	0
B	0
C	0
D	0
E	0
F	0
G	0
H	0
I	0
J	0
K	**0**
L	**0**

Example

Pop the front of the queue



Queue:

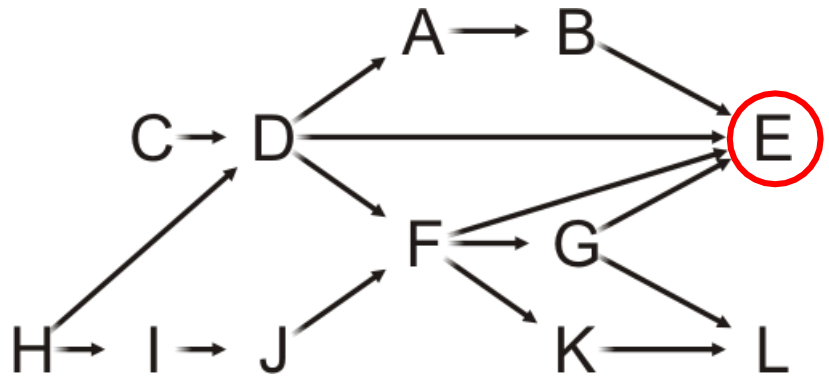
C	H	D	I	A	J	B	F	G	K	E	L

A	0
B	0
C	0
D	0
E	0
F	0
G	0
H	0
I	0
J	0
K	0
L	0

Example

Pop the front of the queue

- E has no neighbors—it is a *sink*



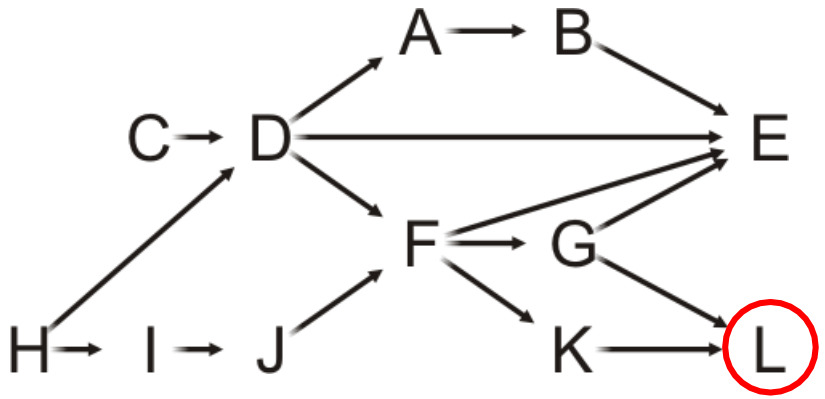
Queue:

C	H	D	I	A	J	B	F	G	K	E	L

A	0
B	0
C	0
D	0
E	**0**
F	0
G	0
H	0
I	0
J	0
K	0
L	0

Example

Pop the front of the queue



Queue:

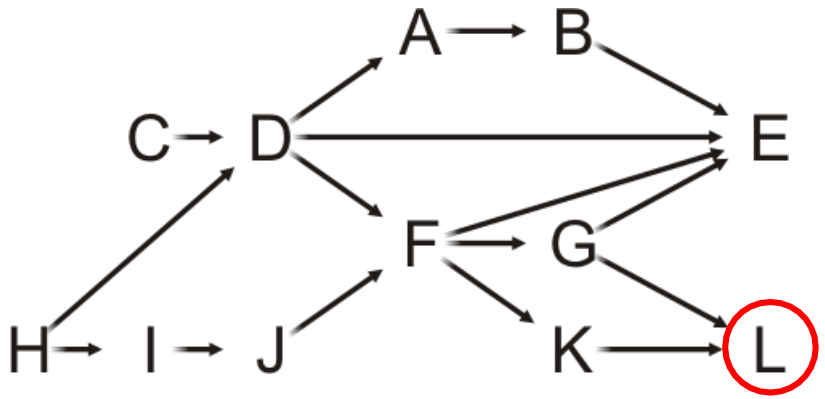
C	H	D	I	A	J	B	F	G	K	E	L

A	0
B	0
C	0
D	0
E	0
F	0
G	0
H	0
I	0
J	0
K	0
L	0

Example

Pop the front of the queue

- L has no neighbors—it is also a *sink*



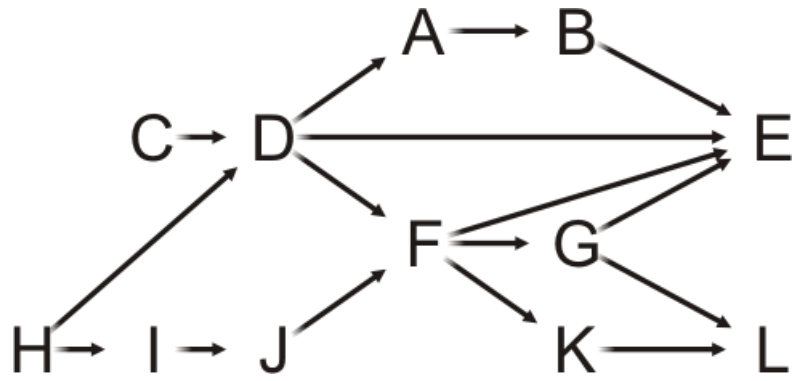
Queue:

C	H	D	I	A	J	B	F	G	K	E	L

A	0
B	0
C	0
D	0
E	0
F	0
G	0
H	0
I	0
J	0
K	0
L	**0**

Example

The queue is empty, so we are done



Queue:

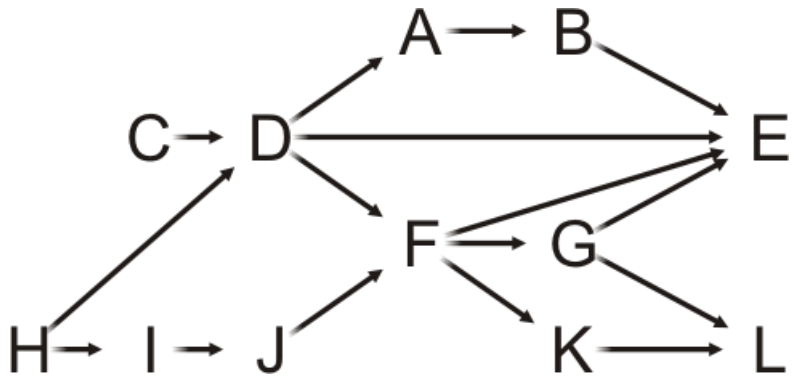
C	H	D	I	A	J	B	F	G	K	E	L

A	0
B	0
C	0
D	0
E	0
F	0
G	0
H	0
I	0
J	0
K	0
L	0

Example

We deallocate the memory for the temporary in-degree array

The array stores the topological sorting



C	H	D	I	A	J	B	F	G	K	E	L

A	0
B	0
C	0
D	0
E	0
F	0
G	0
H	0
I	0
J	0
K	0
L	0

- Time complexity:
  - Calculate the in-degree of each v once: $O(?)$
  - For each vertex, need to consider the associated edges once.
    - $O(?)$: adjacency list
    - $O(?)$: adjacency matrix

Task	If stored using Adj. Matrix	If stored using Adj. List
Initialization: - Allocate memory and **initialize** an **array of in-degrees** - **Initialize *Q*** with all v having in-degree 0	Cost to initialize in-degrees = $O(V^2)$	Cost to initialize = $O(V+E)$
While Q is **not empty**: - **Pop** a vertex from Q - **Decrement** the in-degree of each neighbor - Neighbors whose in-degree **becomes 0 are pushed into *Q***	- For each v, visit the neighbors. - No easy way ! - Check all entries of that row. - Cost of while loop = $O(V \times V) = O(V^2)$	- **Just check dedicated list for that v !** - Each vertex will come in the Queue $O(V)$ - For each vertex, visit the neighbours resulting in $O(E)$ operations in total - So total loop costs $O(V+E)$
	Total cost = $O(V^2)$	Total Cost = $O(V+E)$

- Time complexity:
  - Calculate the in-degree of each v once: $O(V)$
  - For each vertex, need to consider the associated edges once.
    - $O(V+E)$: adjacency list
    - $O(V^2)$: adjacency matrix

- What happens if at some step, all remaining vertices have an in-degree > 0?
  - Consequence: **??**

- What happens if at some step, all remaining vertices have an in-degree > 0?
  - Consequence: we get $O(V + E)$ algorithm for **cycle detection**.

Thank you